How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

Anonymous Authors[1]

Abstract

Prompts for in-context learning (ICL) describe a task, demonstrate it through examples, or both. Yet how descriptions work, especially in relation to demonstrations, remains unclear. Through the Bayesian view of ICL, we characterize descriptions as informative priors by their *specificity* and *reliability*, and adapt the effective sample size (ESS) from Bayesian statistics to quantify their predictive value as an equivalent number of demonstrations. Our theoretical results on in-context linear regression tasks show that a reliable description's ESS grows with its specificity and with the task's dimension. However, the ESS falls sharply if the description may be irrelevant: in a 16-dimensional task with no demonstrations, a one-in-ten chance of irrelevance cuts a precise description's ESS from 117 to 13.5. We then test whether learners attain these values, in experiments with Transformer networks and pretrained language models, including frontier ones. Meta-trained Transformers come close to the Bayes-optimal ESS and adapt to unreliable descriptions when trained on them. Pretrained language models, however, place too little weight on precise descriptions when answering without reasoning, and closely match the Bayesian reference only when reasoning is enabled. Taken together, these results extend the Bayesian analysis of ICL from demonstrations to descriptions.

1. Introduction

Large language models (LLMs) are capable of in-context learning (ICL): given a prompt that specifies a task through a natural-language description, demonstrations, or both, they can predict the output for a new query input (Brown et al., 2020). In practice, prompts consist mostly of system instructions, user instructions, skill files, and earlier conversation turns, which typically describe the task rather than demonstrate it (Mao et al., 2025; Patsakis et al., 2026). The balance shifts further as LLMs carry out increasingly long-horizon tasks (Kwa et al., 2025), for which demonstrations are costly to provide.

Theories of ICL, however, have mostly focused on prompts consisting of demonstrations alone (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023). The few that include descriptions study only specific types, such as a descriptor that describes the input distribution (Huang & Ge, 2025), a set of candidate hypotheses (Lin et al., 2025), or related datasets supplied as context (Zhu et al., 2026); none offers a systematic treatment of descriptions. We therefore ask: how can we characterize natural-language descriptions in general to formally understand and analyze them? And how do descriptions compare, combine, and interact with demonstrations?

We study both questions through the Bayesian view of ICL, in which a model infers a latent task from the demonstrations (Xie et al., 2022); a description can then be modeled as a prior over tasks. Descriptions differ in how far they narrow down the task and in whether they apply: "write this report as an accountant" leaves most of it open, a step-by-step workflow for each part leaves little, and either may be meant for another task. We formalize these two properties in d-dimensional Gaussian linear regression, a standard controlled setting for ICL (Garg et al., 2022; Panwar et al., 2024). Without a description, the coefficients follow a Gaussian baseline prior with variance a_0; a description supplies further information about them before demonstrations are observed. A mixture model separates uncertainty about where the task lies from uncertainty about whether the description applies: with probability p, the description is relevant, and the coefficients are Gaussian around its center with precision c; otherwise, it is independent of the task. We thus characterize a description by two quantities: its *specificity*, set by c, and its *reliability*, the prior probability p that it is relevant.

To compare descriptions with demonstrations, we adapt sample-size measures of prior impact and predictive perfor-

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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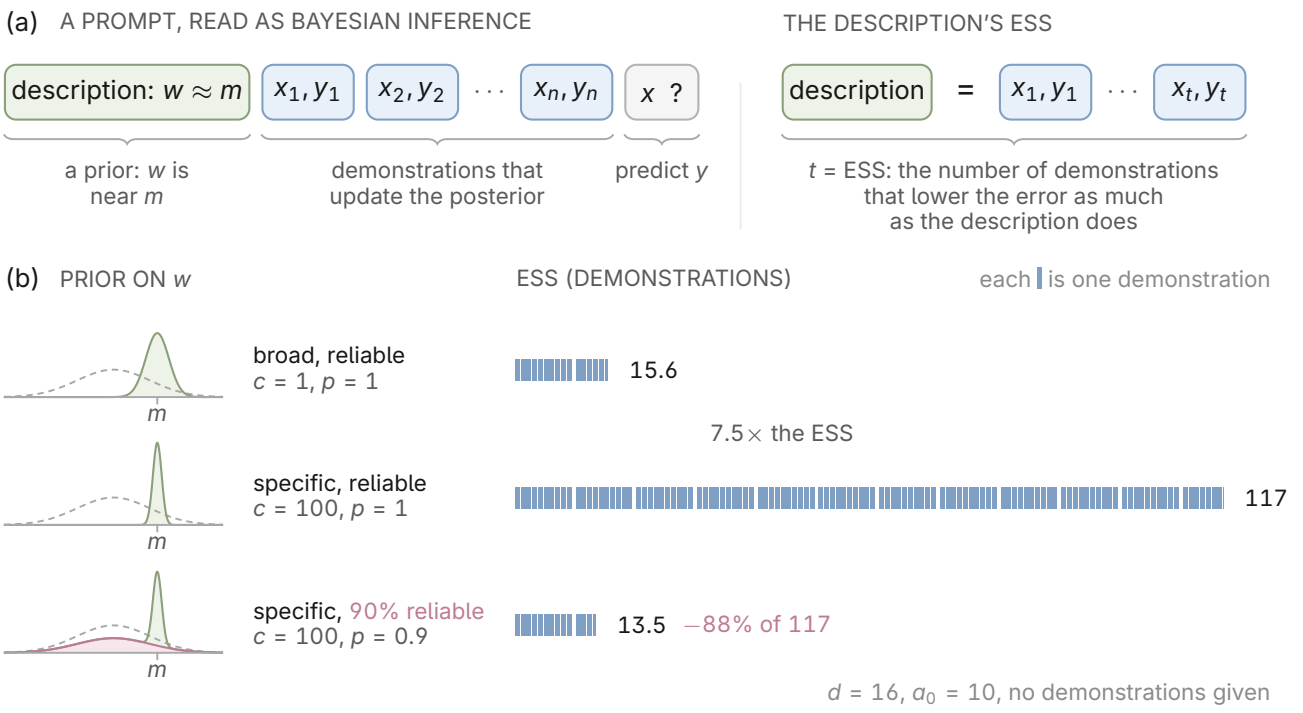


Figure 1. **Measuring a task description in demonstrations.** (a) A description provides information about a latent task w, and demonstrations supply additional evidence. Its predictive effective sample size (ESS) is the number of additional demonstrations a learner without the description needs to match the error it reaches with the description. The schematic on the right illustrates $n = 0$. (b) Description-conditioned task distributions and their Bayes-optimal ESS at $n = 0$, $d = 16$, and $a_0 = 10$. Specificity c is the precision of the relevant component; reliability p is its prior probability. Pink denotes the irrelevant component. Increasing specificity greatly increases ESS when $p = 1$, whereas a 10% probability of irrelevance sharply limits this benefit.

mance (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021; Le Scao & Rush, 2021; Selby et al., 2025). For a given learner and prediction loss, a description's *predictive effective sample size* (ESS) at n demonstrations is the additional number of demonstrations needed without the description to reach the same expected error. Standard prior ESS instead matches information, which requires an explicit prior (Morita et al., 2008); a network or a language model also brings prior knowledge, acquired in training, but holds it only implicitly in its weights, so only its error can be compared. Computing ESS for both the Bayes-optimal predictor and trained models distinguishes the information available under the statistical model from the benefit realized by a learner.

We then ask how closely trained learners attain the Bayes-optimal ESS. Transformers trained from scratch on the Gaussian model receive each description as an input token and must learn its precision and reliability from their training data. Pretrained language models cannot learn these from the task distribution, so the prompt must state them, as probabilistic statements in numerical mean-estimation tasks. Using a stated precision requires weighting the description and the demonstrations by their precisions, a computation that reasoning may help a model carry out, so we compare models with reasoning enabled and disabled.

Our main findings are as follows:

- **Precision and dimension determine the value of reliable descriptions.** With fixed positive integer precision c, zero-demonstration ESS approaches $c + d + 1$ as the baseline prior becomes diffuse (Section 3).
- **Uncertain relevance limits zero-demonstration ESS.** Any fixed positive probability of irrelevance imposes a finite ceiling as precision grows. For $d = 16$, baseline variance $a_0 = 10$, and $c = 100$, lowering reliability from 1 to 0.9 reduces ESS from approximately 117 to 13.5. With demonstrations, a learner can also infer whether the description is relevant (Section 4).
- **Description use depends on training for Transformers, and on reasoning for language models.** Meta-trained Transformers approach the Bayes-optimal ESS, and their fitted description weight follows the reliability of their training data, even when test descriptions are more reliable. For most pretrained language models tested, enabling reasoning brings their use of stated precision close to the Bayesian reference. When descriptions are in fact unreliable and the prompt says so, GPT-6 Sol with reason-

ing also matches the reference, which requires inferring relevance from the demonstrations (Sections 5–6).

2. Framework

Tasks and demonstrations. A task is a vector $w \in \mathbb{R}^d$ with baseline prior $\mathcal{N}(0, a_0 I_d)$, where $a_0 > 0$. A prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query input x, with

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

Inputs and observation noise are independent of the task and one another. Numerical illustrations in Sections 3–4 use $d = 16$ and $a_0 = 10$; the same contrasts hold at other dimensions and baseline variances (Table 1).

Descriptions and relevance. A description supplies a center $m \in \mathbb{R}^d$. Its *specificity* is represented by precision $c \geq a_0^{-1}$, at least that of the baseline prior, and its *reliability* is $p \in [0, 1]$. Let $r = (ca_0)^{-1} \in (0, 1]$ be the fraction of baseline variance remaining in the relevant component. With a latent relevance indicator $Z \sim \text{Bernoulli}(p)$ independent of m, the joint model is

$$\begin{aligned} m &\sim \mathcal{N}(0, (1-r)a_0 I_d), \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d), \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d). \end{aligned}$$

Thus

$$w \mid m \sim p\,\mathcal{N}(m, c^{-1} I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d).$$

This two-component structure is the robust mixture prior that Bayesian clinical trials use to borrow external information that may not apply (Schmidli et al., 2014). Irrelevance is all-or-nothing because a description meant for another task says nothing about this one; a description that is merely inaccurate is a less precise one. Integrating over m recovers $w \sim \mathcal{N}(0, a_0 I_d)$ under either component, so comparisons with and without descriptions concern the same task population.

A description is *relevant* when $Z = 1$ and *irrelevant* when $Z = 0$. We use *reliable* for $p = 1$ and *unreliable* for $p < 1$.

Prediction error. For a prediction $\hat{y}$, we measure expected squared error against the noise-free query response:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{1}$$

We call this quantity *prediction error*, or *risk*; it excludes irreducible query noise. The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, $R^{\text{desc}}(n)$ and $R^{\text{demo}}(n)$ denote risk with and without a description, and $R^{\text{desc}}_{p=1}(n)$ denotes the reliable case. We call R^{demo}, as a function of the demonstration count, the learner's *baseline curve*.

Definition 1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \geq 0$ satisfying $R^{\text{demo}}(t) = R^{\text{desc}}(n)$. When this crossing exists, the description's ESS is

$$\text{ESS}_n = n^* - n.$$

Noninteger counts refer to an interpolated baseline curve.

An ESS of 10 means that n demonstrations plus the description match the error of $n + 10$ demonstrations alone. A learner without a description still has prior knowledge, from the baseline prior or from training, and ESS measures what the description adds to it. ESS is therefore measured against each learner's own baseline curve, so a large ESS need not imply low absolute error, and a negative ESS means that the description raises error.

3. The Value of a Reliable Description

For $p = 1$, a description's value without demonstrations depends on task dimension as well as precision. Write $W_n = X_n^\top X_n$, where the rows of X_n are the demonstration inputs. Gaussian conjugacy gives the Bayes-optimal risks

$$\begin{aligned} R^{\text{demo}}(n) &= \mathbb{E}\,\text{tr}(a_0^{-1} I_d + W_n)^{-1}, \\ R^{\text{desc}}_{p=1}(n) &= \mathbb{E}\,\text{tr}(c I_d + W_n)^{-1}. \end{aligned}$$

We state the result in the limit $a_0 \to \infty$ of a diffuse baseline prior, in which a learner knows little about the task without the description.

Proposition 1 (Reliable descriptions)**.** *Let* $p = 1$.

(i) With $r \in (0, 1]$ *fixed,* $\text{ESS}_0 \geq (1-r)d$*, and* $\text{ESS}_0 \to (1-r)d$ *as* $a_0 \to \infty$.

(ii) With positive integer c *fixed,* $\text{ESS}_0 \to c + d + 1$ *as* $a_0 \to \infty$.

Part (i) holds fixed the fraction $1 - r$ of baseline variance that the description removes, so its precision $c = (ra_0)^{-1}$ decreases as a_0 grows. Before the demonstration inputs span $\mathbb{R}^d$, error in the remaining directions dominates as $a_0 \to \infty$. The description then replaces $(1-r)d$ demonstrations in the limit, with the same quantity providing a lower bound at finite a_0 (Appendix A.1).

Part (ii) holds the precision fixed instead. It follows by comparing the description's risk d/c with the limiting baseline risk $d/(n-d-1)$, valid for integer $n > d + 1$ (Muirhead, 1982): the risks match at $n = c + d + 1$, and for noninteger $c > 1$ the limit lies between those of the neighboring integers. Random inputs constrain some directions less than their expected information $\mathbb{E}W_n = nI_d$ suggests, so $\mathbb{E}\,\text{tr}(W_n^{-1}) = d/(n-d-1)$ exceeds d/n; this difference accounts for the additional $d + 1$ demonstrations.

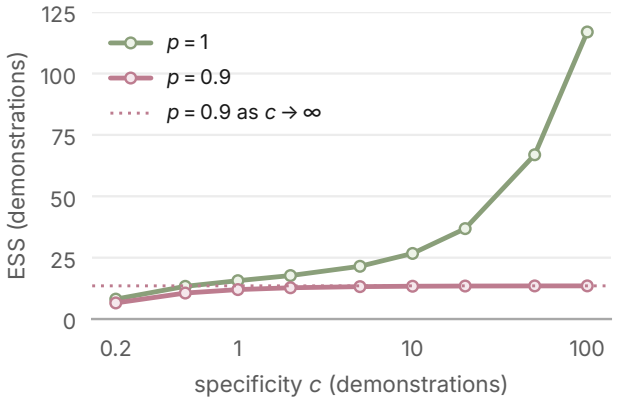


Figure 2. **Reliability limits the benefit of specificity without demonstrations.** Bayes-optimal ESS_0 versus precision c, for $p = 1$ and $p = 0.9$, with $d = 16$ and $a_0 = 10$. The dotted line is the finite ceiling as $c \to \infty$ at $p = 0.9$. Estimates use eight batches of 4,000 prompts; standard errors are below 0.02 demonstrations.

At $d = 16$ and $a_0 = 10$, a reliable description with $c = 1$ has ESS 15.6, compared with the lower bound $(1 - r)d = 14.4$. Increasing c to 100 raises ESS to approximately 117 (Figure 2). With many demonstrations, the dimension-dependent contribution vanishes: $\mathrm{ESS}_n \to c - a_0^{-1}$, the precision that the description adds to the baseline prior, which is its conventional prior ESS (Corollary 3(ii) below). After 100 demonstrations, the $c = 1$ description has ESS 1.1, close to its added precision of 0.9. Descriptions often specify only part of a task; one that constrains only some of its directions can instead gain value as demonstrations accumulate (Appendix A.3).

4. The Cost of Unreliability

We now allow the description to be irrelevant with probability $1 - p$. The Bayes-optimal predictor must then weigh the two components by how probable each is given the demonstrations.

For $0 < p < 1$, let $\mu_z = \mathbb{E}[w \mid m, D_n, Z = z]$ be the component posterior means and let $\pi_n = P(Z = 1 \mid m, D_n)$ be the *posterior relevance probability*. Bayesian model averaging (Hoeting et al., 1999) gives

$$\hat{y} = \big[\pi_n \mu_1 + (1 - \pi_n)\mu_0\big]^\top x.$$

The law of total variance then yields the following risk decomposition.

Proposition 2 (Risk under uncertain relevance)**.** *For every* $n \geq 0$*,*

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1 - p)R^{\mathrm{demo}}(n) + \mathcal{V}_n, \tag{2}$$

where

$$\mathcal{V}_n = \mathbb{E}\Big[\pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2\Big] \geq 0.$$

Without demonstrations,

$$R^{\mathrm{desc}}(0) = \big[1 - p^2(1 - r)\big]a_0 d. \tag{3}$$

Here $\mathcal{V}_n$ is the expected between-component posterior variance. The first two terms in Equation (2) are the risk of the Bayes-optimal predictor that also observes Z, so $\mathcal{V}_n$ is the excess Bayes risk due to unobserved relevance. Without demonstrations, observing Z would remove a fraction $p(1 - r)$ of baseline risk; without Z, the predictor shrinks m by the factor p, and the reduction is only $p^2(1 - r)$ (Appendix A.2).

Corollary 3 (ESS under uncertain relevance)**.**

(i) For $0 < p < 1$*, define* $n_p^\sharp$ *by*

$$R^{\mathrm{demo}}(n_p^\sharp) = (1 - p^2)R^{\mathrm{demo}}(0).$$

Then $\mathrm{ESS}_0 < n_p^\sharp$ *at every finite c, and* $\mathrm{ESS}_0 \to n_p^\sharp$ *as* $c \to \infty$*. As* $a_0 \to \infty$ *with r fixed,* $\mathrm{ESS}_0 \to p^2(1 - r)d$ *and* $n_p^\sharp \to p^2 d$*. For* $p = 0$*, both ESS and the ceiling are zero.*

(ii) With d, a_0*, and c fixed, as* $n \to \infty$*,*

$$\begin{cases} \mathrm{ESS}_n \to c - a_0^{-1}, & p = 1, \\ \limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c - a_0^{-1}), & p < 1. \end{cases}$$

Uncertain relevance imposes an error floor without demonstrations. For fixed $p < 1$, Equation (3) approaches the positive floor $(1 - p^2)a_0 d$ as $c \to \infty$, giving a finite ceiling on zero-demonstration ESS. We take $p = 0.9$, the weight that the robust-prior example of Schmidli et al. (2014) gives its informative component. With $d = 16$ and $a_0 = 10$, increasing c from 1 to 100 then raises ESS only from 12.0 to 13.5, compared with 15.6 to 117 at $p = 1$ (Figure 2). At $a_0 = 10$, the ceiling $n_p^\sharp$ is approximately 13.5, so the $c = 100$ description nearly reaches it; as the baseline prior becomes diffuse, the ceiling tends to $p^2 d = 12.96$.

Demonstrations distinguish low precision from uncertain relevance. Without demonstrations, an unreliable description has the same risk as a reliable description with remaining variance fraction $1 - p^2(1 - r)$. For example, an unreliable description with $c = 100$ and $p = 0.9$ and a reliable one with $c \approx 0.52$ both give ESS about 13.5 here. After 100 demonstrations, their ESS values are 86.6, near the bound $p(c - a_0^{-1}) \approx 89.9$ of Corollary 3(ii), and 0.5. Demonstrations can provide evidence for the high-precision component of the unreliable description; the reliable low-precision description has no such component.

5. Meta-Trained Transformers

Transformers meta-trained on demonstrations can approach Bayes-optimal prediction (Garg et al., 2022; Genewein et al., 2025). We test whether they also attain the Bayes-optimal ESS of a description. Following Garg et al. (2022), we train from scratch on the model of Section 2, with $a_0 = 10$ and, as in Huang & Ge (2025), $d = 5$. A prompt contains an optional description token carrying m, up to 20 demonstration tokens, and a query token (Huang & Ge, 2025). The description token does not state c or p; a network must learn these properties across training tasks. Half the prompts omit the description, which gives each network its own baseline curve. We train separate networks for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$, each a factor of ten apart, with two or three seeds per configuration (Appendix B); a $c = 100$ description is worth more demonstrations than a 20-demonstration prompt can show.

Matched training and test distributions. At $n = 0$, networks trained with $c = 10$ and $p = 1$ achieve mean ESS 14.3, compared with a Bayes-optimal ESS of 16.1 on the same prompts (Figure 3a) and 15.9 on a larger sample (Table 3). At lower reliability, and for $c = 1$, their ESS is within 0.3 of the Bayes-optimal value. The networks also show the dependence on dimension: their ESS rises from 14.3 at $d = 5$ to 17.8 at $d = 10$ (Appendix B).

Reliability shifts. The networks learn reliability from their training data, but descriptions at test time may be more or less reliable. We therefore evaluate each network at test reliabilities that differ from its training reliability, and compare it with two Bayesian predictors: one that uses the training reliability, p_{train}, and the Bayes-optimal predictor, which uses the test reliability, p_{test} (Raventós et al., 2023). At $n = 0$, a Bayesian predictor with reliability p predicts $p\, m^\top x$. We therefore summarize a network's predictions by the *fitted description weight* q, the coefficient from regressing them on $m^\top x$.

The networks keep the reliability they were trained with. For $c = 10$, networks trained at $p = 0.9$ retain weights near 0.9 even when every test description is relevant (Table 3), and so realize only part of the Bayes-optimal ESS (Figure 3b); m alone does not reveal the changed test reliability. The opposite shift, from reliable descriptions in training to unreliable ones at test time, costs little without demonstrations (Figure 3b) but makes the description harmful once six or more demonstrations are observed, which training at $p = 0.99$ avoids (Appendices A.4 and B).

6. Pretrained Language Models

Unlike the meta-trained networks, pretrained language models must be told a description's precision and reliability. We ask whether they use what the prompt states. To isolate the use of a description from the ability to carry out linear regression in context, we use numerical mean estimation, the scalar case with fixed input $x = 1$, which has an exact Bayesian reference.

6.1. Task and Evaluation

Each numerical task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$ and demonstrations drawn from $\mathcal{N}(w, \sigma^2)$, with $\sigma = 60$, then rounded to integers; a task is redrawn if any of its numbers falls outside 100 to 999, because language models predict short unsigned numbers best (Gruver et al., 2023; Requeima et al., 2024). A description gives a center m and a standard deviation τ: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." Before rounding and redrawing, a relevant description satisfies $w \mid m \sim \mathcal{N}(m, \tau^2)$. We measure its precision relative to one observation, $c = \sigma^2/\tau^2$, retaining the unit-noise convention of the theory. Stated standard deviations 60, 30, and 10 correspond to $c = 1$, 4, and 36, so that the reference's weight on the description, $c/(c + n)$, ranges from 0.2 to 0.97 over $n \le 4$; a noise level of $\sigma = 60$ leaves room for this range of scales among three-digit numbers.

Conditions. We vary the stated precision. To isolate the response to it, we also hold m and all demonstrations fixed and describe a center generated at $c = 36$ as having $c = 4$. Section 6.3 adds a stated chance that the description is irrelevant, and Appendix C varies the description's wording.

Models. We select models from different developers, open-weight and proprietary, that can answer with reasoning both enabled and disabled under the same model identifier. The main comparison uses the same 200 tasks for five such models: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026), Gemma 4 31B (Gemma Team, 2026), GPT-6 Sol (OpenAI, 2026), and Claude Opus 4.6 (Anthropic, 2026b). We add another recent frontier model, Claude Haiku 5.5 (Anthropic, 2026a), only with reasoning, at medium effort, because its reasoning could not be disabled in our setup. The three frontier models are served through their developers' command-line agents, with tools disabled. Appendix C gives execution and readout details and a larger run without reasoning on 2,000 tasks.

Bayesian reference. Our *prompt-based Bayesian reference* is the Bayesian predictor for the model that the prompt states: its observation noise, its description precision, and, when given, its chance of irrelevance. Without a description, the prompt says only that the mean is "unknown": a stated distribution for it would itself be a description, and the baseline should reflect only what a model already knows (Section 2). The reference therefore needs a baseline prior

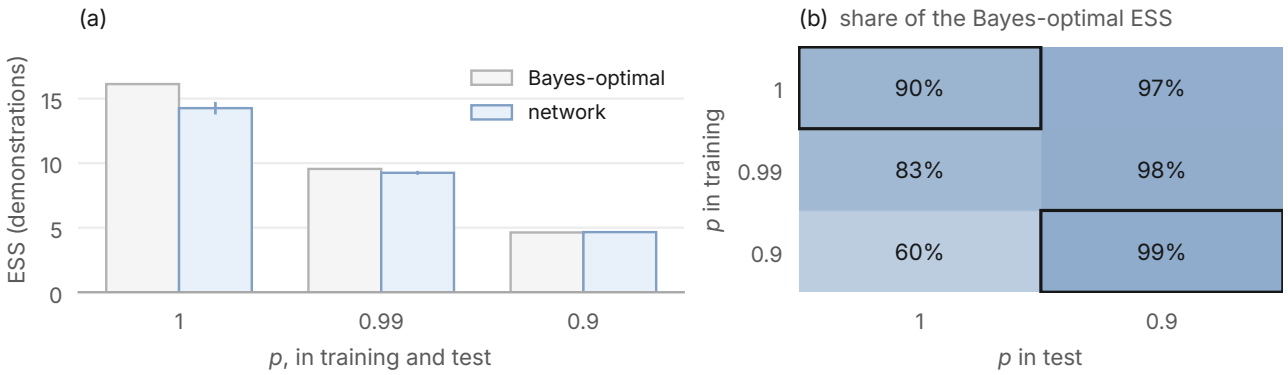


Figure 3. **Meta-trained Transformers approach Bayes-optimal ESS and reflect training reliability.** Results use $c = 10$, $d = 5$, $a_0 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own baseline curve; bars show the mean and one standard deviation across three training seeds. The Bayes-optimal comparison uses the same prompts. (b) Test reliability differs from training reliability. Cells show network ESS as a fraction of the Bayes-optimal ESS under the test distribution; outlined cells have matching reliabilities. Table 3 includes $c = 1$.

of its own, and takes a uniform one over the integers 100 to 999, the range of every number shown in a prompt. With a description, its prediction is approximately the precision-weighted Gaussian posterior mean (Neuenschwander et al., 2020),

$$\hat{w}_{\text{ref}} = \frac{c\,m + n\bar{y}}{c + n}, \tag{4}$$

where $\bar{y}$ is the demonstration mean. With a fixed input there is no dimension term: in the idealized continuous Gaussian model with a flat baseline prior, a correctly specified description has exact ESS c, its conventional prior ESS. The reference follows what the prompt states even when it differs from the generating values, so agreement with it measures how a model uses the stated information.

Measures. We compute ESS against each model's own baseline curve and report *summary ESS*, its median over $n \in \{1, 2, 4\}$: models rarely answer with a number at $n = 0$, and at large n ESS mainly reflects how slowly their baseline curves fall (Appendix C). Applied to the reference, this estimator gives 34.6 at $c = 36$, not exactly 36, because the tasks are rounded and redrawn and the estimator interpolates (Appendix C). We also regress predictions on m, $\bar{y}$, and an intercept. The coefficient on m is the *fitted description weight*; the reference weight is $c/(c + n)$.

6.2. Reasoning and Stated Precision

Precise descriptions are underweighted without reasoning. For $c = 36$, summary ESS ranges from 2.0 to 10.3 across the five models without reasoning, compared with 34.6 for the reference (Figure 4b). The shortfall grows with precision: at $c = 4$, the models reach 0.5 to 2.7, against 3.5 for the reference (Figure 4a), and at $c = 1$, -0.2 to 0.9, against 1.0 (Table 7). Sensitivity to the stated precision itself varies: describing the same center as $c = 4$ instead of $c = 36$ reduces reference ESS from 34.6 to 13.7 (not to 3.5, because the center stays as accurate), GPT-6 Sol's from 10.3 to 2.4, and Opus 4.6's and Gemma 4 31B's by about half; Qwen3.5 9B and gpt-oss-20b change less (Table 7).

Reasoning improves the use of stated precision. With reasoning enabled, Opus 4.6, GPT-6 Sol, and Gemma 4 31B attain approximately the reference ESS at both precisions (Figure 4a,b), and their ESS falls with the reference's when only the stated precision is lowered (Table 7). Haiku 5.5 matches the reference at $c = 4$ and falls short at $c = 36$ (28 against 34.6): it reaches 28 against 33 at $n = 4$, and only 3.4 at $n = 1$, where 28% of its replies contain no recorded reasoning output. The individual replies of these four models follow the reference: on tasks with $|m - \bar{y}| \geq 30$, where the ratio is well determined, 89–100% of replies at $n = 4$ imply a description weight, $(\hat{w} - \bar{y})/(m - \bar{y})$, within 0.05 of the reference weight (Appendix C).

Qwen3.5 9B and gpt-oss-20b also approach the reference weights. Their own baseline curves flatten at larger n and do not fall to their error with the description, so the squares in Figure 4b measure their ESS on the reference's baseline curve. gpt-oss-20b's ESS stays below the reference because about 2% of its replies contain arithmetic errors (Appendix C). Reasoning does not suffice for every model: Olmo 3 7B Think (Team Olmo, 2025), a small open-weight model that always reasons and so has no run without reasoning, reaches summary ESS 6.2 at $c = 36$ (Appendix C).

6.3. Stated Reliability

Section 4 shows that uncertain relevance sharply lowers a description's value, and that demonstrations then carry evidence about relevance. To test both, we make the $c = 36$ description unreliable and say so. The prompt adds that

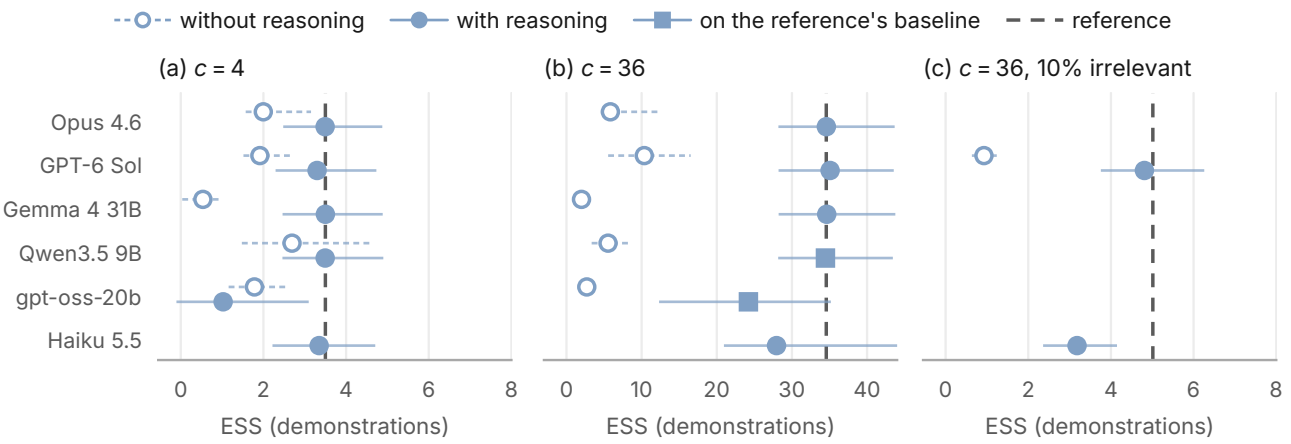


Figure 4. **With reasoning, language models use the precision and the reliability that a prompt states.** Summary ESS on 200 tasks at $\sigma = 60$, with 95% task-bootstrap intervals. Open points indicate reasoning disabled; filled points indicate reasoning enabled; vertical dashed lines show the prompt-based Bayesian reference. (a) Descriptions generated and stated at $c = 4$. (b) Descriptions generated and stated at $c = 36$. (c) The descriptions of (b) when they are irrelevant one time in ten and the prompt says so, for the two models run in this design; each task is evaluated with a relevant and with an irrelevant description. Squares in (b) measure ESS on the reference's baseline curve instead of each model's own, a different comparison. Intervals are clipped to the display.

there is a 10% chance that the statement is wrong, in which case the mean "was instead drawn uniformly at random from the whole numbers 100 to 999," and the stated chance is the true one: each task is run with its relevant description and with an irrelevant one, whose center is drawn independently of the task, and the two errors are averaged with weights 0.9 and 0.1. Because of cost, this design covers two models, GPT-6 Sol and Haiku 5.5; Appendix C.6 also reports a 50% chance.

For the reference, a one-in-ten chance of irrelevance reduces ESS from 34.6 to 5.0 (Figure 4c), as it reduces the Bayes-optimal ESS from 117 to 13.5 in Section 4. Most of what remains comes from the demonstrations: a predictor that keeps the stated 10% whatever they show, which we call *fixed-weight averaging*, has ESS 0.3, because it cannot discount a description that the demonstrations contradict. With reasoning, GPT-6 Sol reaches 4.8, and its replies are within 2 of the reference's prediction in 89% of cases with a relevant description and 92% with an irrelevant one. Haiku 5.5 reaches 3.2, with 85% and 60% of replies within 2 of the reference. Without reasoning, GPT-6 Sol's ESS is 0.9, and at most 19% of its replies are within 2 of the reference.

Updating the chance requires the prompt to say what holds when the description is irrelevant. An earlier version of the warning said only that the mean was then "unknown"; with reasoning, most models then applied the stated chance as given, without updating it (Appendix C.6). The networks take a description's reliability from their training data. With reasoning, GPT-6 Sol takes it from the prompt and, as the reference does, revises it on the demonstrations; Haiku 5.5 does so less consistently.

A semantic task. Descriptions in practice usually state constraints in ordinary language, not distributions. For the same two models, agreement with a Bayesian reference extends to a message-filing task with such descriptions, whose unreliable descriptions are in fact irrelevant in about half of the tasks (Appendix C.8).

7. Related Work

Bayesian accounts of ICL. Theoretical work studies ICL as Bayesian inference over latent tasks (Xie et al., 2022; Panwar et al., 2024), as a learning algorithm implemented in a forward pass (Akyürek et al., 2023; von Oswald et al., 2023), and as meta-learning across tasks (Garg et al., 2022; Müller et al., 2022). Meta-trained predictors can approximate Bayesian inference under suitable training conditions (Genewein et al., 2025). Agreement with Bayesian predictors can depend on training-task diversity and on the function class (Raventós et al., 2023; Panwar et al., 2024), and language-model ICL can violate Bayesian consistency (Falck et al., 2024). Beyond the forms of task information cited in Section 1, theories have studied a prior that mixes task groups (Lin & Lee, 2024) and an instruction whose influence changes (Tong et al., 2026).

Prior effective sample size. Prior ESS expresses the influence of a prior in observation units, with different definitions matching different statistical quantities (Morita et al., 2008; Neuenschwander et al., 2020). Our comparison is closest to measures based on expected loss (Reimherr et al., 2021; Wiesenfarth & Calderazzo, 2020). Le Scao & Rush (2021) measure the average horizontal separation of learn-

ing curves in prompt-based fine-tuning. Selby et al. (2025) compare language-model-elicited priors with data-based predictors by the number of observations needed to match predictive loss. We apply loss matching to task descriptions at each demonstration count, using each learner's own baseline curve.

Descriptions and evidence in language models. Task summaries distilled from demonstrations can replace the examples in context (Honda et al., 2025). Other studies report weak sensitivity to instruction relevance (Webson & Pavlick, 2022) or limited benefits from repository instruction files (Gloaguen et al., 2026). Gupta et al. (2025) compare explicitly stated coin biases with observations and evaluate the predictions against Bayesian predictors. Concurrent work studies how reasoning and code execution affect language models' approximation of Bayesian posteriors (Filbry et al., 2026); our main runs disable code execution. Appendix D discusses further connections.

8. Discussion

We asked how descriptions can be characterized, and how they compare, combine, and interact with demonstrations. Treating a description as a prior with a specificity and a reliability, and measuring it by its predictive ESS, answers both: a reliable description can replace many demonstrations, uncertain relevance imposes a finite ESS ceiling before demonstrations are observed, and demonstrations can then raise a description's ESS by supplying evidence about its relevance. Meta-trained Transformers approach the Bayes-optimal ESS, and their fitted description weight follows the reliability of their training data. Most pretrained language models we test use a stated precision as the Bayesian reference does only with reasoning enabled; when descriptions are unreliable and the prompt says so, GPT-6 Sol with reasoning also matches the reference.

The analysis explains, in a controlled setting, why descriptions sometimes substitute for many examples (Le Scao & Rush, 2021; Honda et al., 2025) and sometimes have limited influence (Webson & Pavlick, 2022; Gupta et al., 2025; Gloaguen et al., 2026). It separates three causes of limited influence: low specificity, uncertain relevance, and a learner's limited use of the available information. More broadly, because a description enters as a prior, the Bayesian analysis of ICL extends from demonstrations to descriptions.

Limitations. Our theory covers one setting: Gaussian linear regression with squared error and a two-component model of irrelevance. Our language-model prompts state a description's precision explicitly, which descriptions in practice rarely do. Under informal wording such as "give or take," several models overweight a vague description without reasoning, and Haiku 5.5 often repeats the description's center even with reasoning (Appendix C); how a model judges a description that leaves its precision unstated remains open. The language-model evidence is also narrow: the tasks are small and synthetic, the frontier models are served through command-line agents (Appendix C), and the test of unreliable descriptions covers two models. Finally, agreement with a Bayesian reference concerns predictions and does not identify a model's internal computation.

Impact Statement

This work studies how learners combine task descriptions with demonstrations. Its measurement framework may inform prompt evaluation and design when descriptions are uncertain or incomplete. The experiments use synthetic tasks; their results do not establish dependable instruction following in consequential applications.

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A. Proofs

A.1. Reliable Descriptions

Notation and risk. Write $\varepsilon = 1/a_0$ for the baseline prior precision. Let $X_n \in \mathbb{R}^{n\times d}$ contain the demonstration inputs as rows, and set $W_n = X_n^\top X_n$. For a prior precision $\kappa > 0$, define

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. Here and throughout, risk means expected squared prediction error against the noise-free query response $w^\top x$. The identities follow from the Gaussian posterior covariance: the posterior mean minimizes conditional squared error, and the independent query satisfies $\mathbb{E}[xx^\top] = I_d$. Averaging the conditional error $x^\top \Sigma x$ therefore gives $\mathbb{E}\,\mathrm{tr}\,\Sigma$.

Throughout the proofs, d is a positive integer, $a_0 > 0$, $c \geq \varepsilon$, $r = 1/(ca_0) \in (0, 1]$, and $p \in [0, 1]$. Specificity is represented by the relevant component's precision c; r is its variance as a fraction of the baseline variance. Reliability p is the prior probability of relevance. Noninteger demonstration counts refer to linear interpolation between the risks at adjacent integers.

Wishart identities. We use three standard properties of W_n.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \geq d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \frac{d}{n-d-1} \quad (n \geq d+2),$$

whereas this expectation is infinite for $n = d$ and $n = d + 1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^\kappa(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing. In the correctly specified joint model, conditioning on the description cannot increase expected posterior variance, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{demo}}(n)$ and the Bayes-optimal ESS is nonnegative.

Proof of Proposition 1. A reliable description without demonstrations has risk

$$R^{\mathrm{desc}}_{p=1}(0) = \frac{d}{c} = ra_0 d.$$

(i) Fixed variance reduction. By (W1), $R^\varepsilon(n) \geq (d-n)a_0$ for $n < d$; the inequality also holds trivially for $n \geq d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{demo}}(t) \geq (d-t)a_0 \qquad (t \geq 0).$$

For $t < (1-r)d$, the right-hand side exceeds $ra_0 d$. The matching sample size must therefore satisfy $\mathrm{ESS}_0 \geq (1-r)d$.

For every integer n,

$$\frac{R^\varepsilon(n)}{a_0} = \mathbb{E}\,\mathrm{tr}(I_d + a_0 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } a_0 \to \infty.$$

Indeed, an eigenvalue $1/(1 + a_0\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real t by interpolation. For fixed $r \in (0, 1)$, the limiting curve crosses rd at the unique point $t = (1-r)d$. Strict monotonicity then gives convergence of the crossing. For $r = 1$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every a_0, giving the same conclusion.

(ii) Fixed precision. For fixed $c > 0$, the description risk d/c is independent of a_0. As $a_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^{\varepsilon}(n) \longrightarrow \begin{cases} +\infty, & n \le d+1, \\ \dfrac{d}{n-d-1}, & n \ge d+2. \end{cases}$$

If c is a positive integer, the finite part of this limiting curve equals d/c at the integer $n = c + d + 1$. For every fixed point below this crossing the limiting risk is larger than d/c, and for every fixed point above it the limiting risk is smaller. The crossing of the interpolated finite-a_0 risk therefore converges to $c + d + 1$. The integer condition ensures that the equality occurs at a grid point; linear interpolation does not in general preserve the same expression for noninteger c. □

Large-sample expansion. For fixed d and κ, commuting W_n with κI_d gives

$$(\kappa I_d + W_n)^{-1} = W_n^{-1} - \kappa W_n^{-2} + \kappa^2 W_n^{-2}(\kappa I_d + W_n)^{-1}.$$

The final term is positive semidefinite and bounded above by $\kappa^2 W_n^{-3}$. Using (W2) and (W3),

$$R^{\kappa}(n) = \frac{d}{n-d-1} - \frac{\kappa d}{n^2} + O(n^{-3}). \tag{5}$$

An additional demonstration reduces the leading term by $d/n^2 + O(n^{-3})$. A fixed increase in prior precision and a fixed increase in the demonstration count therefore reduce risk at the same asymptotic order. This comparison gives the large-sample ESS limits below.

A.2. Unreliable Descriptions

Let $\mathcal{P}_n = (D_n, m, x)$ denote the full prompt, including the query input. The posterior relevance probability is $\pi_n = P(Z = 1 \mid \mathcal{P}_n)$, whereas p remains the prior relevance probability. Conditional on $Z = 1$, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^{\top} y_{1:n}).$$

Conditional on $Z = 0$, the description center m is independent of w, and

$$\Sigma_0 = (\varepsilon I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^{\top} y_{1:n}.$$

The posterior relevance probability follows from the two Gaussian marginal likelihoods,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + c^{-1} X_n X_n^{\top}\big), \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + a_0 X_n X_n^{\top}\big), \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query input does not enter this update because it is independent of (Z, m, D_n). Without demonstrations, both marginal likelihoods equal one and $\pi_0 = p$, since m has the same distribution under both relevance states. At $p = 0$ or $p = 1$, the posterior contains only the corresponding component. At $r = 1$, $m = 0$ almost surely and the two components coincide, so the description provides no additional task information for any p.

Proof of Proposition 2. As in Bayesian model averaging (Hoeting et al., 1999), the posterior mean prediction averages the component predictions using their posterior probabilities:

$$\mathbb{E}[w^{\top} x \mid \mathcal{P}_n] = \big(\pi_n \mu_1 + (1-\pi_n)\mu_0\big)^{\top} x.$$

The law of total variance over Z separates posterior variance into within-component and between-component terms:

$$\begin{aligned} \mathrm{Var}(w^{\top} x \mid \mathcal{P}_n) = \pi_n x^{\top} \Sigma_1 x + (1-\pi_n) x^{\top} \Sigma_0 x \\ + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^{\top} x\big)^2. \end{aligned}$$

For $0 < p < 1$ and any integrable function g of the prompt,

$$\mathbb{E}[\pi_n g(\mathcal{P}_n)] = \mathbb{E}[Z g(\mathcal{P}_n)] = p\,\mathbb{E}[g(\mathcal{P}_n) \mid Z = 1].$$

Conditional on $Z = 1$, the prompt follows the reliable-description model. Conditional on $Z = 0$, the description provides no information about the task. Taking expectations in the variance decomposition therefore gives

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1-p)R^{\mathrm{demo}}(n) + \mathcal{V}_n,$$

with $\mathcal{V}_n$ as in Equation (2). The endpoint cases $p = 0$ and $p = 1$ follow directly from the single-component posteriors.

At $n = 0$, $\mu_1 = m$, $\mu_0 = 0$, and the prediction is $p\,m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = a_0, \qquad \mathrm{Var}(m_i) = (1-r)a_0, \qquad \mathrm{Cov}(w_i, m_i) = p(1-r)a_0.$$

It follows that

$$\mathbb{E}(w_i - pm_i)^2 = a_0 - 2p^2(1-r)a_0 + p^2(1-r)a_0 = \big(1 - p^2(1-r)\big)a_0.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction risk is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (3). □

Proof of Corollary 3. (i) No demonstrations. For $0 < p < 1$ and finite c,

$$R^{\mathrm{desc}}(0) = \big(1 - p^2(1-r)\big)R^{\mathrm{demo}}(0) > (1-p^2)R^{\mathrm{demo}}(0).$$

The risk on the left decreases to the right-hand side as $c \to \infty$, equivalently $r \to 0$. The interpolated baseline risk is continuous, strictly decreasing, and tends to zero. Consequently, the matching sample size is strictly below the unique finite $n_p^\sharp$ defined in the corollary and converges to it as $c \to \infty$. At $p = 0$, the description is always irrelevant, so $\mathrm{ESS}_0 = n_0^\sharp = 0$; the non-strict bound remains valid.

The baseline lower bound and normalized limit established in Appendix A.1 also give

$$\mathrm{ESS}_0 \geq p^2(1-r)d, \qquad \mathrm{ESS}_0 \longrightarrow p^2(1-r)d$$

when $a_0 \to \infty$ with r fixed. Applying the same argument to the level $1 - p^2$ gives $n_p^\sharp \to p^2 d$. These expressions are limits as the baseline prior becomes diffuse; they need not hold exactly at finite a_0.

For comparison, an oracle predictor that observes Z has zero-demonstration risk

$$\big(1 - p(1-r)\big)a_0 d \geq (1-p)a_0 d.$$

The oracle's zero-demonstration ESS ceiling therefore approaches pd as $a_0 \to \infty$. For the learner that must infer relevance, this limit is $p^2 d$. The difference is the cost of uncertainty about relevance.

(ii) Many demonstrations. Fix d, a_0, and c, and define the oracle risk

$$L(n) = pR^c(n) + (1-p)R^\varepsilon(n).$$

Equation (5) implies, uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\left[p(c-\varepsilon) - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0, T]$. Let $n + \eta_n$ be the sample size at which the baseline risk equals $L(n)$. Then

$$\eta_n = p(c-\varepsilon) + O(n^{-1}).$$

When $p(c-\varepsilon) = 0$, the two risks coincide and $\eta_n = 0$ exactly. If $p = 1$, the description risk equals $L(n)$, so

$$\mathrm{ESS}_n \longrightarrow c - \varepsilon = (1-r)c.$$

If $p < 1$, the variance decomposition gives $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$. Since the baseline risk decreases with sample size, $\mathrm{ESS}_n \leq \eta_n$, proving

$$\limsup_{n\to\infty} \mathrm{ESS}_n \leq p(c-\varepsilon) = p(1-r)c.$$

Because $\mathcal{V}_n$ is nonnegative and need not vanish at the required rate, the result for unreliable descriptions is an upper bound. □

Other dimensions and baseline variances. The numerical illustrations in the body use $d = 16$ and $a_0 = 10$. Table 1 repeats the comparison of a reliable and an unreliable $c = 100$ description at other values. In every case the reliable description's zero-demonstration ESS is within 1.5 of $c + d + 1$, and a one-in-ten chance of irrelevance reduces it to near the ceiling $p^2 d$ of Corollary 3.

Table 1. **The cost of unreliability at other dimensions and baseline variances.** Bayes-optimal zero-demonstration ESS of a $c = 100$ description at $p = 1$ and $p = 0.9$, with the limits of Proposition 1(ii) and Corollary 3(i). The baseline curve is estimated from 4,000 sets of demonstration inputs per count (1,500 at $d = 50$).

d	a_0	ESS_0, $p = 1$	$c + d + 1$	ESS_0, $p = 0.9$	$p^2 d$
5	1	105.0	106	8.7	4.05
5	10	105.8	106	4.7	4.05
5	100	105.9	106	4.2	4.05
16	1	115.8	117	17.3	12.96
16	10	116.9	117	13.5	12.96
16	100	117.0	117	13.0	12.96
50	1	149.5	151	44.4	40.50
50	10	150.9	151	40.9	40.50
50	100	151.0	151	40.5	40.50

A.3. Partial Descriptions

Descriptions often specify only part of a task. In our model, a reliable description may constrain only $k < d$ directions. With fixed $c > 0$, its zero-demonstration ESS approaches k as $a_0 \to \infty$: the unconstrained directions dominate the remaining error. With many demonstrations, its ESS instead approaches $k(c - a_0^{-1})/d$ (Proposition 4). A sufficiently precise partial description can therefore gain value as demonstrations resolve what it leaves unspecified. At $d = 16$, $k = 8$, and $c = 100$, ESS rises from 8.1 without demonstrations to 36.2 after 100 demonstrations (Table 2).

Formally, suppose the description leaves a fraction r of the baseline variance in these directions and leaves the remaining $d - k$ directions unchanged. The description-conditioned prior precision is c in the constrained directions and $\varepsilon = 1/a_0$ elsewhere. Write

$$\Lambda = \mathrm{diag}(cI_k, \varepsilon I_{d-k}).$$

The prediction risk is $\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 4 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k < d$ be an integer.*

(i) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1 - r)k$ with r fixed, and $\mathrm{ESS}_0 \to k$ with c fixed.
(ii) For fixed d, a_0, and c, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - 1/a_0) = \frac{k}{d}(1 - r)c.$$

As the baseline prior becomes diffuse with c fixed, the unconstrained directions dominate the zero-demonstration risk, and ESS approaches the number of constrained directions, k. After many demonstrations, ESS instead approaches the added precision averaged across all d directions. A sufficiently precise partial description can therefore be worth more alongside demonstrations than by itself. Table 2 illustrates this complementarity: at $c = 100$, a description of half of the $d = 16$ directions has ESS 8.1 without demonstrations and 36.2 with 100 demonstrations.

Proof of Proposition 4. (i) No demonstrations. The description risk is

$$\mathrm{tr}\,\Lambda^{-1} = (kr + d - k)a_0.$$

Using $R^{\mathrm{demo}}(t) \ge (d - t)a_0$ gives $\mathrm{ESS}_0 \ge (1 - r)k$. The normalized baseline risk converges to $(d - t)^+$, locally uniformly in t, by the integer limits and linear interpolation. With r fixed, the normalized description risk is $d - (1 - r)k$. With c fixed, it is

$$d - k + \frac{k}{ca_0} \longrightarrow d - k.$$

Table 2. **Demonstrations can increase the value of a precise partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $a_0 = 10$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(1 - r)ck/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.05.

	$c = 1$					$c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

Because $k < d$, both limiting levels are positive and have unique crossings, at $(1 - r)k$ and k, respectively. Convergence of the crossings proves the claim. When the description leaves the prior unchanged, its ESS is identically zero.

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E}W_n^{-2} = \frac{\mathbb{E}\,\mathrm{tr}\,W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\mathrm{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\,\mathrm{tr}(MB)| \le \|M\|_{\mathrm{op}}\,\mathrm{tr}\,B$ for $B \succeq 0$, together with $(\Lambda + W_n)^{-1} \preceq W_n^{-1}$. Its absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-1} \le d\|\Lambda\|_{\mathrm{op}}^2\,\mathrm{tr}\,W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n - d - 1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}).$$

Matching this expansion to the baseline risk, as in Corollary 3(ii), gives

$$\mathrm{ESS}_n \longrightarrow \frac{\mathrm{tr}\,\Lambda - d\varepsilon}{d} = \frac{k(c - \varepsilon)}{d}.$$

□

The same expansion applies to any fixed Gaussian description precision matrix $\Lambda \succeq \varepsilon I_d$ compatible with the baseline task distribution. Its large-sample ESS is the added precision averaged across directions, $(\mathrm{tr}\,\Lambda - d\varepsilon)/d$.

A.4. Assuming Relevance Under Misspecification

A learner that has only seen relevant descriptions, such as a network trained at $p = 1$ (Section 5), has no reason to discount one. We ask what assuming relevance costs when descriptions are in fact unreliable. Consider the Bayesian predictor that sets $p = 1$ in its inference model and therefore returns $\mu_1^\top x$. We evaluate its risk, denoted $R^{\mathrm{assume}}(n)$, under the data-generating model with relevance probability p. The assumed prior is misspecified when $p < 1$ and $r < 1$, because it omits the irrelevant component.

Without demonstrations, this predictor returns $m^\top x$. The moments used in the proof of Proposition 2 give

$$R^{\mathrm{assume}}(0) = \mathbb{E}\|w - m\|^2 = \big(1 - (2p - 1)(1 - r)\big)a_0 d.$$

For $r < 1$, this risk exceeds the demonstrations-only risk $a_0 d$ exactly when $p < 1/2$. At $r = 1$, the description carries no information, $m = 0$ almost surely, and the two risks coincide for every p.

With demonstrations, the predictor continues to use $\mu_1^\top x$ regardless of Z, where

$$\mu_1 = m + (cI_d + W_n)^{-1} X_n^\top (y_{1:n} - X_n m).$$

For fixed a_0 and n, $\mu_1 \to m$ in mean square as $c \to \infty$. Conditional on relevance, $\mathbb{E}\|w - m\|^2 = r a_0 d \to 0$; conditional on irrelevance, it is $(2 - r)a_0 d \to 2a_0 d$. Thus

$$\lim_{c \to \infty} R^{\text{assume}}(n) = 2(1 - p)a_0 d.$$

For every $p < 1$, this is a positive risk floor at fixed n, whereas $R^{\text{demo}}(n) \to 0$ as $n \to \infty$. Hence a sufficiently precise description can increase risk relative to the demonstrations-only predictor when relevance is assumed despite $p < 1$. At $p = 0.9$, $c = 100$, $d = 16$, and $a_0 = 10$, the predictor assuming relevance has risk 25.9, compared with 26.6 without the description, at $n = 14$. At $n = 15$, these risks are 25.5 and 19.5, respectively. Its risk remains higher at every larger demonstration count evaluated, up to 10^5. The order of limits matters: the preceding argument takes $c \to \infty$ at fixed n; it does not imply that demonstrations cannot overcome a fixed finite prior precision.

B. Additional Transformer Results and Training Details

Evaluation. We evaluate each network on $20{,}000$ fresh prompts. The networks share these evaluation prompts, so standard deviations across training seeds exclude prompt-sampling variation. At $d = 5$, we evaluate the baseline curve at every integer from 0 to 20, except for $c = 0.2$ and $c = 2$, which use counts from 0 to 15. We compute ESS at the first crossing of this curve with the prediction error obtained using the description, as in Definition 1. Table 3 reports results at $n = 0$, including test reliabilities that differ from the training reliability.

Table 3. **Transformer ESS and fitted description weights under matched and shifted reliability.** Results use $d = 5$, $a_0 = 10$, and no demonstrations, with $20{,}000$ test prompts per network. Rows are grouped by training reliability. *Network* reports mean ESS and standard deviation across the indicated training seeds; for a single seed, the uncertainty is the standard error over prompts. The two Bayesian predictors use p_{train} and p_{test}, respectively; the latter is Bayes-optimal under the test distribution. We show one reference value when they coincide. Seed 0 was not evaluated at $p_{\text{test}} = 0.99$ for training reliabilities 1 and 0.9, so those rows average fewer seeds; their fitted weights differ from the other rows only because of the seeds included. Values using p_{test}, and coincident values using p_{train}, use $200{,}000$ prompts, with standard errors at most 0.1. Other values using p_{train} use the networks' test prompts; sampling variation can change these estimates by up to 0.6, so a Monte Carlo estimate using p_{train} can exceed the corresponding estimate using p_{test}. *Weight* is the fitted description weight q in $\hat{y} \approx q\, m^\top x$, averaged across the available seeds. This coefficient is unrestricted and is not a relevance probability.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Bayes p_{train}	Bayes p_{test}	Weight q
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

Architecture and optimization. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). We train with Adam at learning rate 10^{-4} for 40,000 steps, using batches of 1,024 newly sampled prompts and no curriculum. Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is the posterior mean, which also minimizes squared error. We evaluate the predicted mean using the loss in Equation (1).

Each demonstration token contains an input and its label. Two indicator coordinates distinguish token types, and the model uses no positional encoding. When a description is absent, its token is masked out of attention. The prediction is read from the query token. Labels and description centers are divided by $\sqrt{a_0}$, gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments, which use 0 to 15, and the $d = 10$ experiments, which use 0 to 40.

Layers use pre-normalization, GELU activations, feed-forward width 1,024, and no dropout. At $d = 5$, the model has 9.5 million parameters. Adam uses the PyTorch defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with no weight decay and a constant learning rate. Training seeds are 0, 1, and 2 for $c = 10$, for $d = 10$, and for the partial-description experiments; 0 and 1 for $c = 1$; and 0 for $c \in \{0.2, 2\}$. A $d = 5$ network takes approximately 80 minutes to train on one NVIDIA L4 GPU, and a $d = 10$ network takes approximately 2.8 hours. Training all 25 networks takes approximately 38 GPU-hours.

Without a description, every $d = 5$ network's error is within 0.21 of the Bayes-optimal error of approximately 50 at $n = 0$, and within 0.32 on average over its training demonstration counts. At $d = 10$, the corresponding gaps are at most 0.50 relative to an error of approximately 100, and 0.41. With a description at $c = 10$, $p = 1$, and $n = 0$, network error ranges from 0.67 to 0.71, compared with 0.49 for the Bayes-optimal predictor. The networks therefore remove 98.6% of the baseline error, compared with 99.0% for the Bayes-optimal predictor.

Additional specificity levels. Four additional networks cover $c = 0.2$ and $c = 2$, each at $p = 1$ and $p = 0.9$. Their zero-demonstration ESS values are within 0.03 of the Bayes-optimal values evaluated on the same prompts: 2.64 versus 2.66, 2.16 versus 2.15, 7.32 versus 7.35, and 4.38 versus 4.37, respectively.

Higher dimension and partial descriptions. At $d = 10$, we train three seeds with description precision $c = 10$ and extend training and evaluation to 40 demonstrations so that the baseline curve reaches the ESS crossing. Network ESS is 17.8, compared with a Bayes-optimal value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$, the networks capture most of the description's predictive value but fall short of the Bayes-optimal ESS.

We also train networks with a partial description, as defined in Appendix A.3. The description token contains m in the first $k = 3$ of the $d = 5$ coordinates and zeros elsewhere; the remaining task coordinates follow the baseline prior. For this $c = 10$ description, mean ESS across three seeds increases from 3.27 at $n = 0$ to 5.55 at $n = 10$. The corresponding Bayes-optimal values are 3.22 and 5.64. This increase agrees with the behavior described by Proposition 4.

ESS with demonstrations. Figure 5 evaluates each network at its training reliability. For $c = 1$, network ESS remains within 0.6 of the Bayes-optimal value at every demonstration count whose crossing lies within the evaluation grid. For $c = 10$, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. At $n = 5$, the three $c = 10$ networks trained at $p = 0.9$ have ESS values of 4.83, 4.79, and 4.77, compared with a Bayes-optimal value of 5.20 on the same prompts.

Reliability shift after demonstrations are observed. For $c = 10$, networks trained at $p = 1$ and $p = 0.99$ both have fitted description weights near one at $n = 0$. With demonstrations, their responses to a shift in test reliability differ. At test reliability $p = 0.9$, the description reduces error for networks trained at $p = 0.99$ at every evaluated n, as it does for the Bayes-optimal predictor under the test distribution. For networks trained at $p = 1$, the description instead increases error from six demonstrations onward in every seed. The excess error ranges from 0.77 to 1.06 at $n = 6$ and peaks between 2.74 and 2.94 at $n = 9$. A Bayesian predictor that assumes relevance ($p = 1$) also first has higher error than the demonstrations-only predictor at $n = 6$, with excess error 1.12 there and 2.96 at $n = 9$ (Appendix A.4), so these networks behave as if every description were relevant.

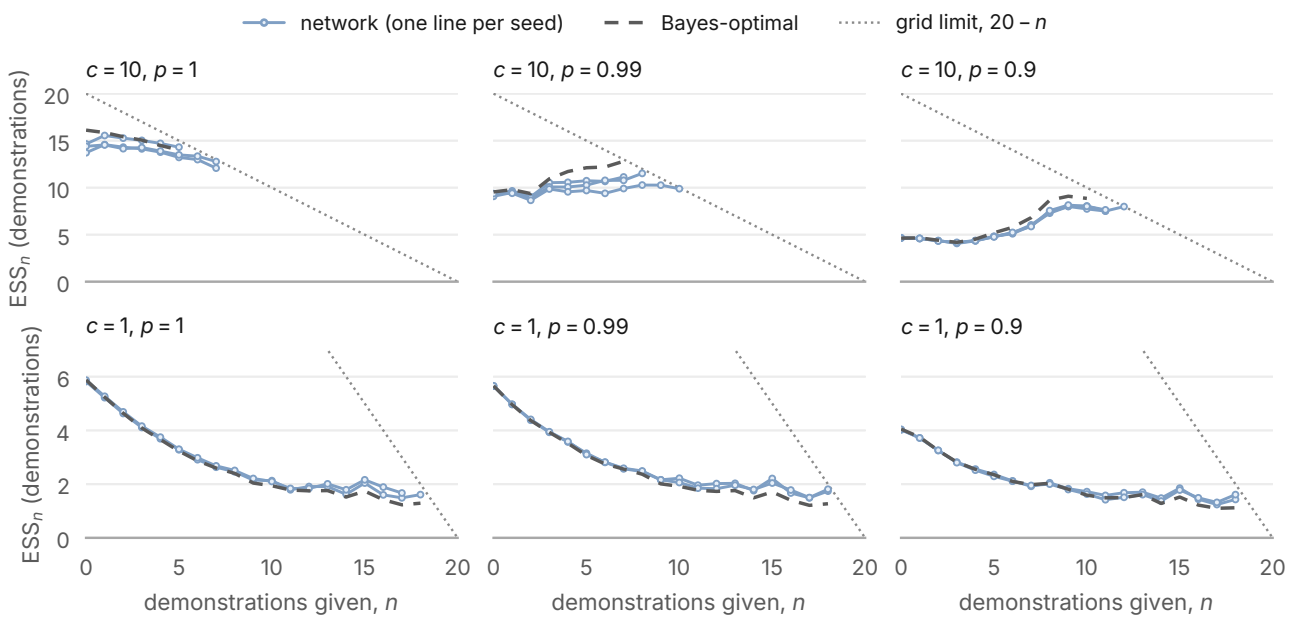


Figure 5. **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayes-optimal predictor is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

C. Additional Language-Model Results and Evaluation Details

To estimate behavior without reasoning more precisely and across more models, we generate 2,000 tasks. Six open-weight models answer all tasks without reasoning; we refer to these experiments as the *full run*. The models are gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B, which also appear in Section 6, together with Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. The main numerical experiments in Section 6 use the first 200 of these tasks. The semantic experiment uses a separate set of 200 tasks, described in Appendix C.8. Unless stated otherwise, the results in this appendix use the full run without reasoning.

Table 4 reports summary ESS, defined as the median over $n \in \{1, 2, 4\}$, with uncertainty intervals. Figure 6 shows the underlying learning curves. Tables 12 and 13 report ESS at every evaluated demonstration count; Table 5 compares interpolation methods; and Table 14 reports description weights for the models with a distribution readout.

For the numerical experiments, the *prompt-based Bayesian reference* combines the probabilistic statements in the prompt with an assumed baseline prior when the prompt does not specify one. It is optimal under that assumed model, but need not minimize error under the actual task distribution when the two differ. We therefore distinguish this reference from the correctly specified Bayesian predictor in the theory. The semantic experiment specifies a finite task distribution and uses its exact Bayesian predictor.

In the numerical task, $c = \sigma^2/\tau^2$ is the prior precision expressed in units of one demonstration. We distinguish the precision used to generate a description value from the precision stated in the prompt whenever they differ. Similarly, a stated reliability supplies a prior relevance probability to the model; it need not equal the proportion of relevant descriptions in the evaluation sample.

C.1. Prompts, Tasks, and Model Readouts

Model inclusion. We retain a model only if it produces a number in at least three quarters of replies overall and, averaged over $n = 1$, 2, and 4 and both noise levels, in at least half of replies without a description. The latter responses determine the baseline curve used for summary ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or "Given…" despite the numeric-only instruction.

The full run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are sampled by a seeded generator. Reasoning is disabled where the chat template supports this option. For gpt-oss-20b, generation begins in the final channel, after the channel normally used for reasoning.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution, renormalized over the 900 three-digit number tokens. We call this the *distribution readout*. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Table 4 for gpt-oss-20b, and Table 6). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Table 4. **Description ESS and weights in the full run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or accompanied by a stated 10% or 50% chance of being wrong. Brackets give 95% bootstrap intervals. The censoring marker $\leq$ indicates that one demonstration alone already outperforms the description-conditioned prediction at a boundary used in the summary. Right: the coefficient on the stated value at $n = 1$; the Bayesian reference weight is $c/(c + n)$. All results use $\sigma = 60$. "Written" denotes the written readout.

	ESS, median over $n = 1, 2, 4$						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Reference	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
548
431
Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." In the rerun that states the baseline prior, the final clause becomes "in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description value is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. A task is redrawn if a displayed number or description value falls outside 100 to 999; this occurs for less than 1% of tasks. The full run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table 12).

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 6, 12, 13, and 14 report this quantity as "number". Tables 6 and 14 italicize cells whose rate is below three quarters.

Written-readout implementation and checks. The written readout extracts the number in the last `\boxed{}` expression in the final answer, if present. Otherwise, it takes the last number with three digits before an optional decimal part, retaining

the decimals. A digit group after a thousands separator, as in 14,945, does not count as a qualifying number. In the full run, decoding is greedy with a limit of eight new tokens, so the reply is the number itself. Those predictions were recorded using an earlier readout that took the first qualifying number and retained only its integer part. Replies without a qualifying number are excluded from both error curves and weight regressions; the reported response rates identify the extent of this exclusion.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout summary ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 58% of written replies across all conditions, so Table 4 reports only its distribution readout. In the wording experiment, however, its numeric response rate exceeds three quarters for the original wording. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

C.2. ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 1 to squared error against the hidden mean. In the full run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. We interpolate squared error, rather than its square root, linearly in $\log n$. The ESS at n is the first interpolated demonstration count t that matches the description-conditioned error, minus n. For a decreasing baseline curve, a negative ESS indicates higher error with the description than without it at the same demonstration count. For the Bayes-optimal predictor under the correctly specified model, conditioning on a description cannot increase expected risk, so its ESS is nonnegative.

If the description-conditioned error lies below the entire baseline curve up to $t_{\max}$, we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we use the marker $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. These markers record censoring at the evaluation boundaries; they do not define an extrapolated matching count or an exact ESS when the crossing does not exist. Summary ESS uses $n = 1$, 2, and 4. We exclude $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error without a description ranges from 9 to 20, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve.

Bootstrap replicates resample tasks jointly for the description-conditioned and baseline curves and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below. We use 500 replicates with a fixed seed and the same resampled tasks for every model.

Reference prior and exact sample equivalence. The prompt-based Bayesian reference takes "unknown" to mean uniform over 100 to 999. Without a description, its prediction is therefore approximately the demonstration mean, with negligible boundary truncation for the evaluated prompts. For a relevant Gaussian description with correctly stated precision, the idealized continuous model with a flat baseline prior gives exact ESS c. Integer rounding, finite-range truncation, and the task-rejection rule are not part of that exact identity; we compute the reported reference values on the same generated tasks as the models. When a chance of being wrong is stated, the reference uses it as the prior probability of irrelevance and computes the posterior relevance probability from the demonstrations. Otherwise, it treats the description as relevant.

Applying the empirical ESS procedure to the full-run reference gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the 200 tasks of Section 6, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6. With a stated chance of irrelevance, the reference's summary ESS varies more across task samples: on 20,000 tasks it is 7.3 at a stated 50% and 23.4 at 10%, compared with 5.8 and 22 on the full run's 2,000 tasks and 5.7 and 27.5 on the first 200. Each model is compared with the reference on the same tasks.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and 35.64. This is the scalar analogue of $c - a_0^{-1}$ in Section 3. We use the prompt-based reference because the models are not given the actual task

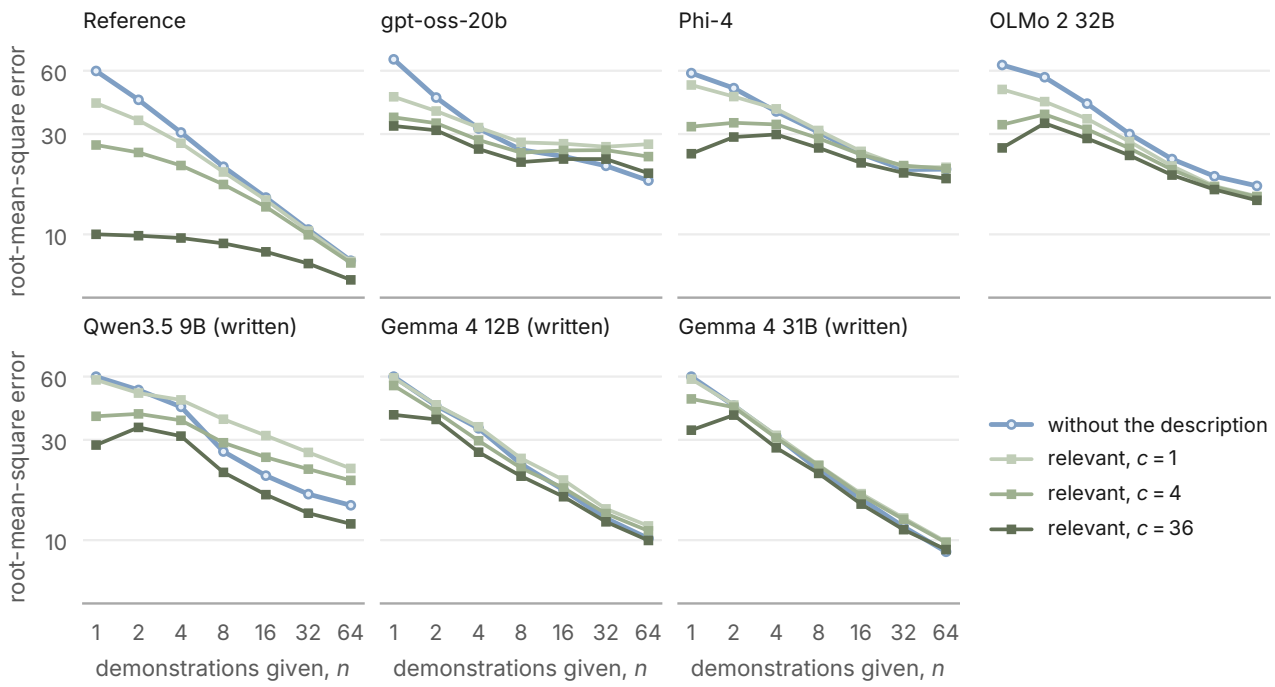


Figure 6. **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the full run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the description-conditioned point at n to the baseline curve at the same error. For four models, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does (Table 14).

prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Sections 3–4, a reference that incorrectly assumes an irrelevant description to be relevant can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

Sensitivity to interpolation. The full-run grid doubles n, so interpolation can affect ESS. On the reference's idealized errors, σ^2/n without a description and $\sigma^2/(c+n)$ with one, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, the relevant crossings are unique. Among the twelve baseline curves across the two noise levels, only Phi-4 at $\sigma = 60$ increases, slightly, from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 5 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using the pool-adjacent-violators algorithm; and a relative-least-squares fit of $A/(n + n_0) + B$ to the baseline errors. The final form is a Bayesian learning curve plus an error floor and recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's summary ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and $c = 36$ has ESS 17 under the estimator in Table 12, but 9.2 under the fitted curve.

C.3. Irrelevant Descriptions and Wording

The main experiments keep every description relevant and state its precision in one wording. This section relaxes each of these choices in the full run.

Irrelevant descriptions. An irrelevant description uses an independently drawn value. At $c = 36$, these descriptions have negative ESS for every model and for the prompt-based reference, which treats them as relevant. The models assign less weight to the description and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range

Table 5. **Sensitivity of summary ESS to baseline-curve estimation.** Median ESS over $n = 1$, 2, and 4 for relevant descriptions, using the full run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Reference	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

from 35 to 64, compared with 124 for the reference.

Wording without reasoning. Descriptions in practice express uncertainty in many ways. To test whether the results depend on the original wording, we repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 6 uses the full-run tasks and written readouts.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. A fitted coefficient above 1 means that, holding the demonstration mean fixed in the regression, a unit change in the description value corresponds to more than a unit change in the prediction. Nevertheless, none of the tested wordings yields the reference's sensitivity to prior precision: across models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this comparison also varies how precisely the task is specified.

C.4. Reasoning Runs and Execution Settings

Comparisons between modes. The labels "with reasoning" and "without reasoning" refer to generation settings. Statements about individual replies refer to recorded reasoning output. Paired runs use the same model identifier and prompts, but reasoning settings, serving software, readout, retry policy, and access to calculation tools vary across the runs. These comparisons concern the evaluated configurations and do not identify a unique internal mechanism.

Shared evaluation prompts. Most configurations in Figure 4 are evaluated on the same 7,800 prompts from the first 200 tasks. These include no-description prompts at $n = 1$, 2, 4, . . ., 64, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b and Olmo 3 7B Think (Team Olmo, 2025) runs omit the two additional baseline points and contain 7,400 prompts. Results without reasoning for the open-weight models are taken from the full run restricted to these tasks. Paired mode comparisons use the same prompts, although serving environments differ as detailed below. At least 99.7% of replies in each run contain a number.

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 repeat themselves until they reach the token limit. Each is rerun once with the model card's recommended reasoning sampling settings: temperature 1.0, top-p 0.95, top-k 20, and presence penalty 1.5. Eleven remain unfinished and one request fails.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun with the model card's recommended sampling settings: temperature 1.0, top-p 0.95, and top-k 64.

Olmo 3 7B Think (`allenai/Olmo-3-7B-Think`) runs in vLLM with greedy decoding and at most 32,768 new tokens.

Table 6. **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the full-run tasks. "Number" denotes the numeric response rate averaged over demonstration counts; a weight is italicized when the response rate at that count is below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. Wording changes overall reliance on the description, but the observed change with stated precision remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

We rerun the 169 replies that reach this limit once with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model generates 9,215 tokens on average and writes a worked solution before most answers. Its long greedy outputs are sensitive to hardware: an earlier stated-precision run on a different GPU produces the same final number on only half the prompts. We therefore report one run. It averages the demonstrations about as well as the reference but places more weight on the description at $c = 1$ and 4. At $c = 36$, its replies split between the description value alone and other prediction rules, as detailed below.

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`, the model identifier served by Codex during the runs) runs through the Codex command-line agent in a read-only sandbox, with an empty configuration directory so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate the model with reasoning disabled (no reasoning tokens) and enabled at low effort, at which its pilot answers were already within one of the reference.

Claude Opus 4.6 (`claude-opus-4-6`) and Haiku 5.5 (`claude-haiku-5-5`) run through the Claude Code command line, with one prompt per call, no tools, no project or user instructions, and the system prompt "You are a helpful assistant." Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled, so that the two runs differ only in reasoning; the disabled mode emits no reasoning tokens. With reasoning enabled, at least 99% of replies in each description condition include reasoning output, with an average of 599 reasoning tokens across all replies. Without a description, only 1%, 24%, and 79% of replies include reasoning output at $n = 1$, 2, and 4, respectively. Haiku 5.5 is evaluated at medium effort, because at low effort it often answers without reasoning, and at low effort for comparison. At medium effort and $c = 36$, its summary ESS is its value at $n = 4$, 28 against 33 for the reference; its ESS at $n = 2$ matches the reference (35), and at $n = 1$ it is 3.4, where 28% of its replies contain no recorded reasoning output (19% and 20% at $c = 1$ and $c = 4$). Both Claude models use the command line's default sampling settings, which it does not report, and one reply per prompt. Without reasoning, Opus 4.6 still writes out intermediate steps in 64 of its 7,800 replies; accounting for

Table 7. **Description ESS across reasoning configurations.** Results use the first 200 tasks at $\sigma = 60$. Summary ESS uses each run's own baseline curve; parentheses report the value computed using the reference baseline where it differs. "Stated irrelevance" gives the stated probability that a description generated at $c = 36$ is irrelevant. The last column fixes the generating precision at 36 and states it as 4. Each run names the mode and, in parentheses, the provider's effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 denotes a crossing beyond the grid. Reference values use the same tasks and estimator. The open-weight rows without reasoning take these 200 tasks from the full run, whose results on all 2,000 tasks are in Table 4. The runs also include three further fixed-value conditions, not shown.

					Stated irrelevance		
Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10%	50%	Stated $c = 4$ Generated $c = 36$
Reference	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.8	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, without reasoning	0.0 (-0.2)	0.5 (0.2)	2.0 (1.6)	≤ -0.3	1.0 (0.7)	0.8 (0.4)	1.0 (0.6)
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, without reasoning	-0.2 (-0.8)	2.7 (0.4)	5.5 (1.1)	≤ -1.0	1.6 (-0.7)	2.3 (-0.6)	4.1 (1.1)
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	29 (27)	3.9	15 (14)
gpt-oss-20b, without reasoning	0.5 (0.3)	1.8 (1.3)	2.7 (1.7)	≤ -0.5	1.8 (1.5)	1.7 (1.5)	2.5 (1.6)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	8.5 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.2 (3.3)	24	≤ -1.0	27 (26)	5.5 (5.6)	8.6 (8.4)
Olmo 3 7B Think, with reasoning	-0.5	0.0	6.2 (6.0)	≤ -1.0	5.9 (5.6)	5.5 (5.3)	15 (13)

these replies changes no summary ESS by more than 0.05.

Run dates. The full run took place on 2–3 October 2026, and its control conditions and gpt-oss-20b's reasoning run on 20 October. Haiku 5.5 was evaluated on 20 October for the main runs and 4 November for wording, the stated baseline prior, and the semantic task. Opus 4.6, Qwen3.5 9B, Gemma 4 31B, and Olmo 3 7B Think were evaluated on 3–4 November. GPT-6 Sol was evaluated on 4 November. The GPT-6 Sol logs do not record a model version beyond the identifier above.

ESS when baseline arithmetic fails. With reasoning, gpt-oss-20b and Qwen3.5 9B nearly match the reference description weight: at $n = 4$ and $c = 36$, their fitted weights are 0.89 and 0.90, compared with 0.90 for the reference. Their errors without descriptions stop falling at larger n. gpt-oss-20b makes errors when averaging long sequences: its no-description root-mean-square error is 19.0 at $n = 32$ and 24.7 at $n = 64$, compared with reference errors 10.0 and 7.4. At these counts, 37 and 62 replies, respectively, differ from the sample mean by more than 20. Qwen3.5 9B levels off near 11 from $n = 32$ onward (11.2 at $n = 64$), with up to 11 replies per cell that far from the sample mean.

For these models, some description-conditioned errors never cross the baseline curve within the evaluated grid. Table 7 therefore also reports, in parentheses, a comparison using the reference's baseline. Specifically, it finds the reference demonstration count t_{ref} that matches the model's description-conditioned error and reports $t_{\text{ref}} - n$. This comparison differs from learner-specific ESS because it uses another predictor's baseline curve. gpt-oss-20b also makes arithmetic errors with descriptions: about 2% of its replies are more than 40 from the reference prediction, from errors such as exchanging the two variances. These errors reduce its ESS; for example, its summary ESS is 1.3 when the prompt states a 10% chance of irrelevance.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks. For these two models, changing between the model and reference baselines changes ESS by at most 1. At low effort and $n = 1$, Haiku 5.5 often repeats the single demonstration value without reasoning output. Its description weight at $n = 1$ is consequently low even in conditions whose ESS is close to the reference value.

C.5. Prediction Rules and Relevance Inference

Per-response description weights. At $n = 4$, we compute the per-response description weight $(\hat{w} - \bar{y})/(m - \bar{y})$. This quantity describes one response and differs from the fitted coefficient obtained by regressing predictions across tasks. We use the tasks with $|m - \bar{y}| \geq 30$, 61 at $c = 36$ and 124 at $c = 1$, and exclude replies without a number. A reply is classified as consistent with a rule when its per-response weight is within 0.05 of that rule's weight. The candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with its two precisions exchanged (0.10 and 0.80); the description counted as one additional observation, $1/(n + 1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 match the reference rule on 89 to 100% of eligible replies. Without reasoning, 77% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Olmo 3 7B Think averages demonstrations well but reaches summary ESS only 6.2 at $c = 36$. At $n = 1$, it repeats the description value in 57 to 80% of replies, more often at higher stated precision. At $n = 4$, about half of its eligible replies return the description value alone and a third match the reference. Without a description, its root-mean-square error falls from 56 at $n = 1$ to 10.8 at $n = 32$, compared with 10.0 for the reference. Its shortfall therefore lies in combining the description with demonstrations. Enabling reasoning does not ensure agreement with the reference in every model.

C.6. Stated Reliability

We test stated reliability in two designs. In the matched design of Section 6.3, descriptions are irrelevant with the stated probability and the prompt states what holds in that case. In an earlier design, every description remained relevant and the warning left the irrelevant case open; we report it second, because it shows what models do when the prompt does not determine how to update.

Matched design. The prompt adds the sentence "There is a 10% chance that this statement is wrong, in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999," or the same sentence with 50%. Each of the 200 tasks is evaluated with its relevant $c = 36$ description and with an irrelevant one, whose center is drawn independently of the task from the same marginal distribution. With stated chance $1 - p$, the error with the description is p times the mean squared error with the relevant description plus $1 - p$ times that with the irrelevant one, so that the stated chance is the true one. ESS is then computed as in the main runs, against each model's own baseline curve, with intervals that resample tasks. *Fixed-weight averaging* conditions each component on the demonstrations and averages the two posterior means with the prior weights p and $1 - p$; a predictor that ignores the warning treats every description as relevant.

Table 8 reports the results. At a 10% chance, the reference's summary ESS is 5.0, against 34.6 without the warning, and the two simpler predictors have ESS 0.3 and -0.1: the description's remaining value depends on recognizing, from the demonstrations, the tasks on which it is irrelevant. At a 50% chance, the reference's ESS is 0.9, and both simpler predictors do worse with the description than without it. GPT-6 Sol with reasoning is within 0.2 of the reference at both chances, and about nine in ten of its replies are within 2 of the reference's prediction with either kind of description. Haiku 5.5 falls short, at a 10% chance mainly with irrelevant descriptions and at 50% with both kinds. Without reasoning, GPT-6 Sol gains little or nothing from the description.

Table 8. **Unreliable descriptions with the stated chance of irrelevance.** Summary ESS in the matched design, with 95% task-bootstrap intervals, and the share of replies within 2 of the reference's prediction when the description is relevant (rel.) and irrelevant (irr.), pooled over $n = 1$, 2, and 4. A value of ≤ -1.0 is censored: at $n = 2$ the description does worse than one demonstration alone.

	10% irrelevant			50% irrelevant		
	ESS	rel.	irr.	ESS	rel.	irr.
Reference	$5.0\ [3.9, 6.2]$			$0.9\ [0.5, 1.2]$		
Fixed weights	$0.3\ [-0.2, 1.0]$			$-0.7\ [-1.0, -0.4]$		
Ignores the warning	$-0.1\ [-0.6, 0.6]$			≤ -1.0		
GPT-6 Sol, low effort	$4.8\ [3.8, 6.3]$	89%	92%	$0.8\ [0.5, 1.2]$	90%	92%
Haiku 5.5, medium effort	$3.2\ [2.4, 4.1]$	85%	60%	$0.6\ [0.3, 0.9]$	58%	52%
GPT-6 Sol, without reasoning	$0.9\ [0.6, 1.2]$	19%	12%	$-0.1\ [-0.2, 0.1]$	10%	12%

When the prompt leaves the irrelevant case open. In the earlier design, the warning gave a 10% or 50% chance that the description is "wrong," in which case "the mean is unknown," and the descriptions themselves remained relevant. A stated 50% chance of irrelevance then reduces reference ESS from 34.6 to 5.7 under the uniform baseline prior (Table 7). Without reasoning, ESS falls by 36 to 62% for four models, against 84% for the reference; GPT-6 Sol's falls from 10.3 to 0.9.

Fixed-weight averaging. The Bayesian reference averages its two component predictions with the posterior relevance probability π_n, which the demonstrations move away from the stated p. For $c = 36$, we compare it with *fixed-weight averaging*, a predictor that conditions each component on the demonstrations and averages the resulting posterior means with the prior weights p and $1 - p$. In the scalar approximation, this gives weight $p\,36/(36 + n)$ to m. Table 9 reports fitted description weights under the original warning, which leaves the baseline prior unspecified. At the stated 50%, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B with reasoning are close to fixed-weight averaging. Opus 4.6 and Haiku 5.5 at medium effort fall below the Bayesian reference weight by $n = 4$. Haiku 5.5 at low effort comes closest to the reference; its replies often contain no reasoning output at $n = 1$. The individual replies show the same pattern. Pooling $n = 1$, 2, and 4 at the stated 50%, the share of replies within 2 of the fixed-weight prediction is 100% for Gemma 4 31B, 97% for Qwen3.5 9B, 89% for GPT-6 Sol, 80% for Opus 4.6, 58% for Haiku 5.5 at medium effort, and 21% for gpt-oss-20b. The share within 2 of the reference prediction is at most 24% for every model. Regressing the replies' normalized description weights on the reference's π_n, as described below, gives slopes of 0.01 for GPT-6 Sol, 0.02 for Gemma 4 31B, 0.06 for Qwen3.5 9B, 0.22 for Haiku 5.5, 0.30 for Opus 4.6, and 0.00 for gpt-oss-20b, where updating as the reference does gives 1.

Table 9. **Description weights under stated unreliability.** Fitted coefficients on the $c = 36$ description value for the first 200 tasks, with stated irrelevance probability $1 - p$. All model rows use reasoning. "Fixed weights" uses weight $p\,36/(36 + n)$; the reference updates the relevance probability using a uniform baseline prior over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. At the stated 50%, several reasoning configurations closely track fixed-weight averaging.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.49	0.46	0.43	0.88	0.84	0.80
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

Stating the baseline prior. Computing π_n requires the marginal likelihood of the demonstrations under each component, and hence the baseline prior under irrelevance. The original warning says only that the mean is otherwise "unknown," so the prompt does not determine a unique posterior relevance probability: a model must supply the prior from its pretrained knowledge, and the reference must assume one. The assumption matters: with its baseline curve held fixed, the reference's summary ESS at the stated 50% is 5.7 under our uniform prior over 100 to 999, 4.6 under the generating distribution $\mathcal{N}(500, 100^2)$, and 13.4 under a uniform prior over 0 to 2000. To remove this ambiguity, we rerun the stated-reliability condition on the same 200 tasks, at $c = 36$, stated irrelevance probabilities of 10% and 50%, and $n = 1$, 2, and 4. The prompt now states the reference's uniform baseline prior over 100 to 999. We use GPT-6 Sol at low effort and Haiku 5.5 at medium effort, with 1,200 prompts each; these runs are also the relevant descriptions of the matched design above. Table 10 compares their fitted description weights with those from the original runs.

At the stated 50%, the reference and fixed-weight predictions differ by at least 5 on 123 to 155 tasks per cell. On these tasks, the share of replies closer to the reference than to the fixed-weight prediction rises from 0–5% to 92–97% for GPT-6 Sol, and from 21–41% to 87–95% for Haiku 5.5. At the stated 10%, at most 7 tasks per cell separate the two rules by this amount, so the comparison is uninformative there. Pooling the three values of n at the stated 50%, the share of replies within 2 of the reference prediction rises from 10% to 90% for GPT-6 Sol and from 23% to 58% for Haiku 5.5.

A constant increase in the description weight could also make replies closer to the reference, because its posterior relevance probability exceeds the stated 50% on most tasks. To test whether the weights vary with the evidence, we use each reply's

normalized description weight: its per-response weight divided by $c/(c+n)$, restricting to tasks with $|m - \bar{y}| \geq 30$. For the scalar reference, this equals its posterior relevance probability π_n, up to the negligible boundary effects discussed above. For model predictions, the ratio can lie outside $[0, 1]$ and does not identify an internal probability. Here π_n ranges from 0.08 to 0.88 as the demonstrations provide different evidence for relevance; fixed-weight averaging keeps the normalized weight constant.

We regress the models' normalized weights on π_n, pooling $n = 1$, 2, and 4 at the stated 50% (269 replies per model). With the baseline prior stated, the slopes are 0.96 [0.84, 1.04] for GPT-6 Sol and 0.98 [0.82, 1.15] for Haiku 5.5, compared with 0.01 [-0.09, 0.11] and 0.22 [-0.01, 0.45] under the original warning. The 95% intervals resample the 153 tasks. The corresponding intercepts with the prior stated are 0.01 and 0.03, and R^2 is 0.57 and 0.33.

The agreement also holds when the demonstrations count against the description. Among the 23 replies with π_n below 0.5 (mean 0.33), the mean normalized weights are 0.30 and 0.38 for the two models with the baseline prior stated, compared with 0.48 and 0.50 under the original warning. Both models therefore vary their description weights across tasks in approximately the amount predicted by Bayesian model averaging.

Summary ESS uses each model's baseline curve from the main run. At the stated 50%, fixed-weight averaging gives 4.4 and the Bayesian reference gives 5.7. Stating the baseline prior raises ESS from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5. At the stated 10%, it changes from 29.7 to 25.0 for GPT-6 Sol and from 32.6 to 15.1 for Haiku 5.5, compared with 27.5 for the reference. The two models' fitted weights agree within 0.01 in this condition (Table 10). Haiku's lower ESS therefore does not reflect lower average reliance on the description; it comes from a minority of replies far from the reference, some of which repeat a demonstration value. Two further configurations were run with the baseline prior stated. Haiku 5.5 at low effort also updates (slope 0.93), with summary ESS 2.7 at the stated 50%, against 5.5 under the original warning. Opus 4.6 without reasoning does not (slope -0.18), with summary ESS 3.8 under both warnings.

Table 10. **Stating the baseline prior changes description weights.** Fitted coefficients on the $c = 36$ description value for the first 200 tasks, as in Table 9. "Unstated" denotes the original warning, "the mean is unknown"; "stated" specifies that the mean is otherwise uniform over 100 to 999, as the reference assumes. GPT-6 Sol uses low effort and Haiku 5.5 uses medium effort.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Fixed weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

C.7. Wording with Reasoning Enabled

Because reasoning brings description weights close to the reference under the original wording, we test whether it does so under the alternative wordings as well. We evaluate the three alternative wordings with Haiku 5.5 at medium effort, for the three relevant descriptions at $n = 1$, 2, and 4 on the first 200 tasks; all $5{,}400$ replies contain a number. With "An earlier estimate put its mean at m, with a standard error of τ", at least 71% of replies in each cell include reasoning output; 99% of these lie within 1 of the reference prediction. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", reasoning output is absent from 4 to 36% and 2 to 74% of replies per cell, respectively. Even among replies with reasoning output, 71% and 91% repeat the description value, while 31% and 20% lie within 1 of the reference prediction. At $c = 1$ and $n = 4$, its description weights are 0.72 and 0.93, compared with reference weight 0.20.

All wordings provide a value m and a scale τ, but "give or take" and "$\pm$" do not identify τ as a prior standard deviation or specify a distribution for the mean. The comparison varies the explicitness of the probabilistic specification as well as its wording. The results show that accurate combination of descriptions and demonstrations under explicit statistical statements need not extend to these less explicit formulations.

C.8. A Semantic Task

The numerical descriptions state a distribution over the task, whereas descriptions in practice usually state constraints in ordinary language that the model must interpret. We therefore design a message-filing task with such descriptions that still admits an exact Bayes-optimal predictor. As in Bayesian models of rule-based concept learning (Goodman et al., 2008), the hidden task is one of a finite set of rules, here 60 filing policies under a uniform prior. A policy note constrains the policy in ordinary language: a broad note leaves 20 compatible policies, and a specific note leaves 2. With reasoning, Haiku 5.5 and GPT-6 Sol obtain ESS 9.9 and 9.5 for the specific note at $n = 2$, compared with 11.4 for the reference, and when the note may be irrelevant, most of their predictions are closer to the Bayesian reference than to fixed-weight averaging.

Task. A message has six binary properties: its sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property as present or absent, using one of three phrasings for each value, such as "I'm a contractor from another company." or "I'm on the sales team here." Properties are sampled independently with probability one half, phrasings are sampled uniformly, and sentences appear in random order.

A policy uses exactly two properties and requires each to be present or absent. A message goes into folder A when both requirements hold, and into B otherwise. There are $\binom{6}{2}2^2 = 60$ such policies. The prompt describes this family and states that the company's policy was chosen uniformly from it. It then provides a policy note, n previously filed messages with their folders, and four new messages. The model is asked for each new message's probability of going into A, as a number between 0 and 1, on four lines from "1: <probability>" to "4: <probability>". Without reasoning, GPT-6 Sol identifies each property correctly in 97–100% of 96 messages; all three misreadings concern one phrasing, "I'm not traveling for this."

Policy notes. A broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," leaving 20 possible policies. A specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," leaving 2. These notes communicate constraints in ordinary language; the model must determine which policies satisfy them.

The unreliable note adds "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." In the generator, the note is relevant with probability one half and otherwise describes an independently drawn policy. The irrelevant component occurs in 92 of the 200 tasks. An independently generated note can agree with the actual policy by chance. Thus, *relevance* denotes the latent generating component, whereas *compatibility* denotes whether a policy satisfies the note; the two are not interchangeable.

Reference and ESS. For a note asserted to be relevant, the reference predicts the fraction of policies consistent with the note and the filed messages that assign the new message to A. For an unreliable note, it averages this prediction with the prediction from policies consistent with the filed messages alone, weighting the two by the posterior relevance probability. The prior relevance probability is 0.5; the posterior probability is obtained using the likelihood of the demonstrations under each component. On 3,000 simulated tasks, the reference's predictive probabilities are calibrated in every condition, and its posterior probability that the unreliable note is relevant matches the frequency of relevance: where it exceeds 0.7, it averages 0.80, and the note is relevant in 80% of cases.

The loss is the squared error between the predicted probability of A and the true folder indicator, averaged over the four query messages. ESS follows Definition 1, using each learner's baseline curve, its loss without a note, at $n = 0$, 2, 4, 6, 8, 12, 16, 20, 24, and 32, with linear interpolation in n. On simulated tasks, this grid changes the reference ESS by at most 0.6 relative to evaluation at every integer demonstration count.

Runs. We use 200 tasks, each defined by a sampled policy, and 19 conditions: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4. This gives 3,800 prompts per configuration. Haiku 5.5 runs at medium effort; GPT-6 Sol runs without reasoning and at low effort. The command-line agents follow Appendix C, with no tools, and no reply calls a tool. Fourteen replies from GPT-6 Sol at low effort contain no probability and are excluded; one Haiku 5.5 reply omits one of its four probabilities. Every Haiku 5.5 reply includes reasoning output, with a median of 3,008 reasoning tokens. For GPT-6 Sol at low effort, 91% of replies include reasoning output, with a median of 142 reasoning tokens.

Results. Table 11 reports each note's ESS. With reasoning, both models are within 2 demonstrations of the reference in every condition except GPT-6 Sol's specific note at $n = 4$: its ESS is 26.1, compared with 14.8 for the reference. Its loss

with this note, 0.008, is close to the reference's 0.007, but its baseline curve flattens beyond $n = 16$, with loss 0.022 at $n = 20$, compared with 0.004 for the reference. It consequently needs more demonstrations on its own baseline curve to match the note's error reduction.

For the unreliable note, we compare predictions with the Bayesian reference and with fixed-weight averaging, which averages the two component predictions using their prior weights. Here it retains the stated 50% weight unless the filed messages rule out every policy compatible with the note; in that case, it assigns zero weight to the note-conditioned component. We use query messages on which the two predictions differ by at least 0.05: about 150 at $n = 2$ and 200 at $n = 4$, out of 800. The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5; 77% and 87% for GPT-6 Sol at low effort; and 45% at both counts for GPT-6 Sol without reasoning.

Without reasoning, GPT-6 Sol learns slowly from demonstrations. Its loss without a note is still 0.078 at $n = 32$, compared with 0.000 for the reference at the reported precision. Measured against this weaker baseline, a note can have an ESS as large as the reference's or larger. For example, the specific note has ESS 13.5 at $n = 2$, although the model's loss with that note is 0.130, compared with 0.027 for the reference. This illustrates why learner-specific ESS must be interpreted alongside absolute prediction error.

Table 11. **ESS of policy notes in the semantic task.** Results use 200 tasks and each learner's own baseline curve, measured without a note. "≤ 0" means that the note did not lower loss; "> 28" means that the baseline curve does not reach the note-conditioned loss by $n = 32$. On $6{,}000$ simulated tasks, the reference ESS is 9.6 and 13.2 for the specific note at $n = 2$ and 4, and 2.5 and 2.9 for the unreliable note. Models are compared with the reference on the same 200 tasks. Paired 95% bootstrap intervals over tasks ($1{,}000$ replicates) for the specific note at $n = 2$ are: reference [9.7, 12.9], Haiku 5.5 [8.7, 11.5], GPT-6 Sol at low effort [7.8, 12.7], and GPT-6 Sol without reasoning [3.7, 21.9].

Note	n	Reference	Haiku 5.5 medium	GPT-6 Sol low	GPT-6 Sol no reasoning
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

C.9. Complete Tables

Tables 12–14 report ESS and description weights at every evaluated demonstration count for the full run without reasoning.

Table 12. **ESS by description condition and demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. The censoring marker $\leq$ marks error above the first baseline point; $> 64 - n$ means that the description-conditioned error lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Reference	$\sigma=60, c=1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	>0
	$\sigma=60, c=1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma=60, c=4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	>0
	$\sigma=60, c=4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma=60, c=36$, relevant		36	36	37	38	39	>32	>0
	$\sigma=60, c=36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma=30, c=1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	>0
	$\sigma=30, c=4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	>0
	$\sigma=30, c=36$, relevant		42	43	43	43	42	>32	>0
	$\sigma=60, c=4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	>0
	$\sigma=60, c=4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma=60, c=36$, relevant, 10% wrong		13	22	26	35	38	>32	>0
	$\sigma=60, c=36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma=60, c=4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	>0
	$\sigma=60, c=4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma=60, c=36$, relevant, 50% wrong		3.1	5.8	12	21	31	>32	>0
	$\sigma=60, c=36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma=60, c=1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma=60, c=4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma=60, c=4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma=60, c=36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma=60, c=36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma=30, c=1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma=30, c=4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma=30, c=36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	>0
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma=60, c=1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma=60, c=1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma=60, c=4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma=60, c=4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma=60, c=36$, relevant	0.86	15	7.6	4.7	5.6	8.5	>32	>0
	$\sigma=60, c=36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma=30, c=1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma=30, c=4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma=30, c=36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	>0
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	>32	>0
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.2	>0
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	>48	>32	>0
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.7	>0
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	>48	>32	>0
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	>32	>0
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	9.2	9.7	11	>56	>48	>32	>0
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	>32	>0
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	>32	>0
	$\sigma=60, c=1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	>0
	$\sigma=60, c=4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	>32	>0
	$\sigma=60, c=4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	>0
	$\sigma=60, c=36$, relevant	0.91	11	4.6	5.3	6.8	15	>32	>0
	$\sigma=60, c=36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	>0
	$\sigma=30, c=1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	>0
	$\sigma=30, c=4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	>32	>0
	$\sigma=30, c=36$, relevant	0.93	7.0	4.4	4.0	5.9	14	>32	>0
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	>0
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	>0
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	>0
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	>0
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	>0
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	>0
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	>32	>0
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	>0

Table 13. **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the full run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table 12.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Qwen3.5 9B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma = 60, c = 4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma = 60, c = 36$, relevant	1.00	6.5	4.0	2.8	6.9	17	> 32	> 0
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma = 30, c = 4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma = 30, c = 36$, relevant	1.00	4.7	4.3	3.0	9.2	14	> 32	> 0
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma = 60, c = 4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma = 60, c = 36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	> 0
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma = 30, c = 4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma = 30, c = 36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	> 0
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma = 60, c = 1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma = 60, c = 1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma = 60, c = 4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma = 60, c = 4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma = 60, c = 36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma = 60, c = 36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma = 30, c = 1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma = 30, c = 4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma = 30, c = 36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma = 60, c = 4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma = 60, c = 4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma = 60, c = 36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma = 60, c = 36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma = 60, c = 4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma = 60, c = 4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma = 60, c = 36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma = 60, c = 36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table 14. **Description weights by demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is defined as in Table 12; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Reference	$\sigma=60, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60, c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60, c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60, c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60, c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60, c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60, c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60, c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60, c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60, c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60, c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30, c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30, c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30, c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60, c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60, c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60, c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60, c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60, c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60, c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30, c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30, c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30, c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma=60, c=1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma=60, c=4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma=60, c=4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma=60, c=36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma=60, c=36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma=30, c=1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma=30, c=4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma=30, c=36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01

D. Additional Related Work

Task descriptions and demonstrations. Weller et al. (2020) formalize learning from task descriptions by distinguishing the latent task, its natural-language description, and individual input–output examples. We use *task description* for information about the task supplied in the prompt, including a numeric description token in the meta-trained setting and probabilistic statements or policy constraints in the language-model experiments. A *demonstration* is an observed input–output example; in scalar mean estimation, it is an observation at the fixed input $x = 1$. Summaries distilled from demonstrations can serve as descriptions (Honda et al., 2025); our controlled experiments instead specify the joint distribution of task, description, and demonstrations.

Prior information and uncertain relevance. The two-component construction is related to robust mixture priors, which combine informative and baseline distributions and update their component weights after observing data (Schmidli et al., 2014). Our mixture is specified directly, rather than derived from a historical-data meta-analysis. Reliability p is the prior probability of the description-conditioned component; π_n is its posterior probability. Averaging component predictions using π_n is Bayesian model averaging (Hoeting et al., 1999). In a robust mixture prior, the prior weight is fixed but the posterior weights still update; fixed-weight averaging, our experimental comparison, omits this update.

In our model, an irrelevant description is independent of the task and may happen to agree with it. Irrelevance is therefore a property of how the description was generated, whereas prior–data conflict concerns disagreement between prior information and observed data.

Specificity and description weights. Specificity takes a concrete form in each setting: the prior precision c in the Gaussian model, together with the number k of constrained directions for a partial description, and the number of compatible policies in the semantic experiment. These quantities do not share a scale across tasks; ESS expresses each in demonstrations. Our description weights follow Gaussian mean estimation, in which the posterior combines the prior mean and the sample mean with weights proportional to their precisions (Neuenschwander et al., 2020). Fitted and per-response weights summarize model predictions; their normalized values can be compared with posterior component probabilities but need not be probabilities.

Effective sample size and predictive comparisons. Prior ESS has several definitions, including matching local information (Morita et al., 2008), matching expected posterior loss (Reimherr et al., 2021), and quantifying sample savings at a target estimation accuracy (Wiesenfarth & Calderazzo, 2020). Matching information requires an explicit prior. A network or a language model brings prior knowledge from training, but that knowledge is implicit in its weights, whereas its prediction error can be measured with and without the description. Our definition therefore follows the risk-matching approach, but it evaluates population prediction error, allows a non-Bayesian learner, and depends on the demonstration count. The word *predictive* specifies what is matched; it does not assert the additive predictive-consistency property studied by Neuenschwander et al. (2020).

Two predictive comparisons are especially close. Le Scao & Rush (2021) average horizontal data advantages over an accuracy range in prompt-based fine-tuning; we evaluate a pointwise difference in demonstration counts. Selby et al. (2025) compare priors elicited from language models with data-based predictors by the observations needed to match predictive loss; we study descriptions supplied to a learner and how their benefit changes after demonstrations. Beyond ICL, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026). Wenliang et al. (2026) compare optimized prompts with demonstrations in terms of their length and effect.

Demonstrations, text, and probabilistic prediction. Other work examines semantic priors and flipped labels (Wei et al., 2023), weak dependence on demonstration labels (Min et al., 2022), Bayesian scaling laws (Arora et al., 2025), and Bayesian reference curves for in-context function learning (Akata et al., 2026). Text supplied alongside numbers can change numerical predictions (Requeima et al., 2024). Related work studies how language-model outputs compare with Bayesian inference and how training can improve probabilistic reasoning (Qiu et al., 2026); He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions. Our numerical experiments state uncertainty explicitly, allowing comparison with a specified Bayesian reference as precision and relevance probability vary. The semantic task extends this comparison to ordinary-language constraints on a finite family of classification policies.